

Drug Delivery: Engineering and Principles

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Bioengineering (BE)

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Drug Delivery: Principles and Applications

- BE 210 (AUG) 3:0
- TA: Kiran Rao U (kiranu@iisc.ac.in) and Subrahmanyam B R (subrahmanya6@iisc.ac.in)
- Generally, I would be available after classes
- Do drop me an email if you plan to come to my office to meet.
- Course Eligibility:
 - All graduate students
 - Undergraduates who have completed their 2nd year and counts as assorted credits

Drug Delivery: Principles and Applications

- Prerequisites: None. Some background in biology is preferred but not required
- All lecture videos are available freely (and transcribed) through NPTEL platform
 - <https://nptel.ac.in/courses/102/108/102108077/>
- References (not required):
 - Drug Delivery: Engineering Principles for Drug Therapy by Mark Saltzman
 - Drug Delivery: Fundamentals and Applications, Anya M. Hillery and Kinam Park, 2nd Edition, CRC Press, 2016

Grading Scheme

- Class participation: 28 marks
 - Announced quizzes (2 to 4): 10-15 marks
 - If you missed a quiz due to medical or other valid reason, then either you can be given the average of the quiz for the whole class, or I can take your viva separately
 - Class participation: 10-15 marks
 - Attendance will be taken in a few classes (quiz and group projects).
- Punctual
- Exam (end of November): 30 marks
- Group Project (presentation by mid Nov): 45 marks
 - May require a separate 2-3 h session outside of the class schedule
- Grading: Will curve

Course Outline

- Introduce concepts of drug delivery for medical challenges
- Pharmacokinetics: Bioavailability, Elimination, Therapeutic index
- Prodrugs, Controlled release
- Polymers: Synthesis, properties, characterization, crystallinity and amorphousness
- Biopolymers: Natural and Synthetic, biocompatibility, Biodegradation, commonly used biopolymers
- Polymer-Drug conjugates, PEGylation
- Diffusion controlled systems, Ficks laws, Reservoir systems, Non-erodible matrix systems, Bio-erodible Systems

Course Outline

- Hydrogels: Physical or chemical, in-situ crosslinking
- Nano and Micro-particles: Dendrimers, Liposomes, Micelles
- Metal and polymeric particles, effect of particle shape, charge and elasticity
- Protein Adsorption and tissue engineering, Drug delivery in tissue engineering
- Implant associated infections, Route specific delivery: Oral, Subcutaneous, Intramuscular, transdermal, inhalation, intravenous
- Vaccines, Cancer vaccines, Cell and gene delivery, Smart responsive drug delivery, Targeted drug delivery, Nanotoxicology and market translation

Course Outline

- Key papers will be discussed throughout the course
- Group project: Propose a project with drug delivery application for an existing medical need
 - Clear need
 - Survey of latest literature
 - Strategy
 - Pros and cons

Drug Delivery

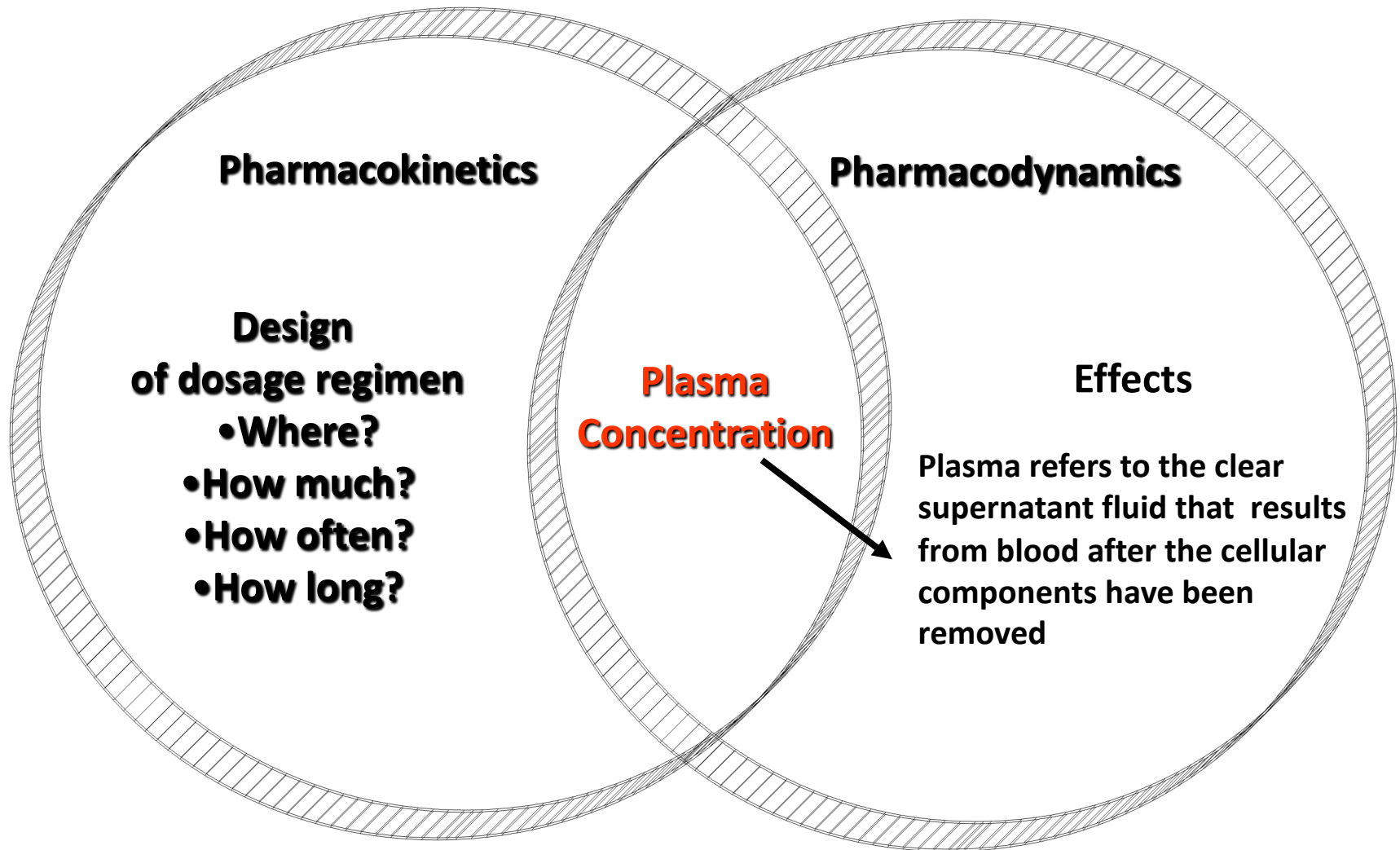
- The appropriate administration of drugs through various routes in the body for the purpose of improving health
 - All diseases need drugs to be delivered
 - Highly interdisciplinary
 - Biology
 - Physiology
 - Materials
 - Engineering
 - It has recently evolved to take into consideration
 - Drug physico-chemical properties
 - Body effects and interactions
 - Improvement of drug effect
- Eventual goal is patient comfort and well being

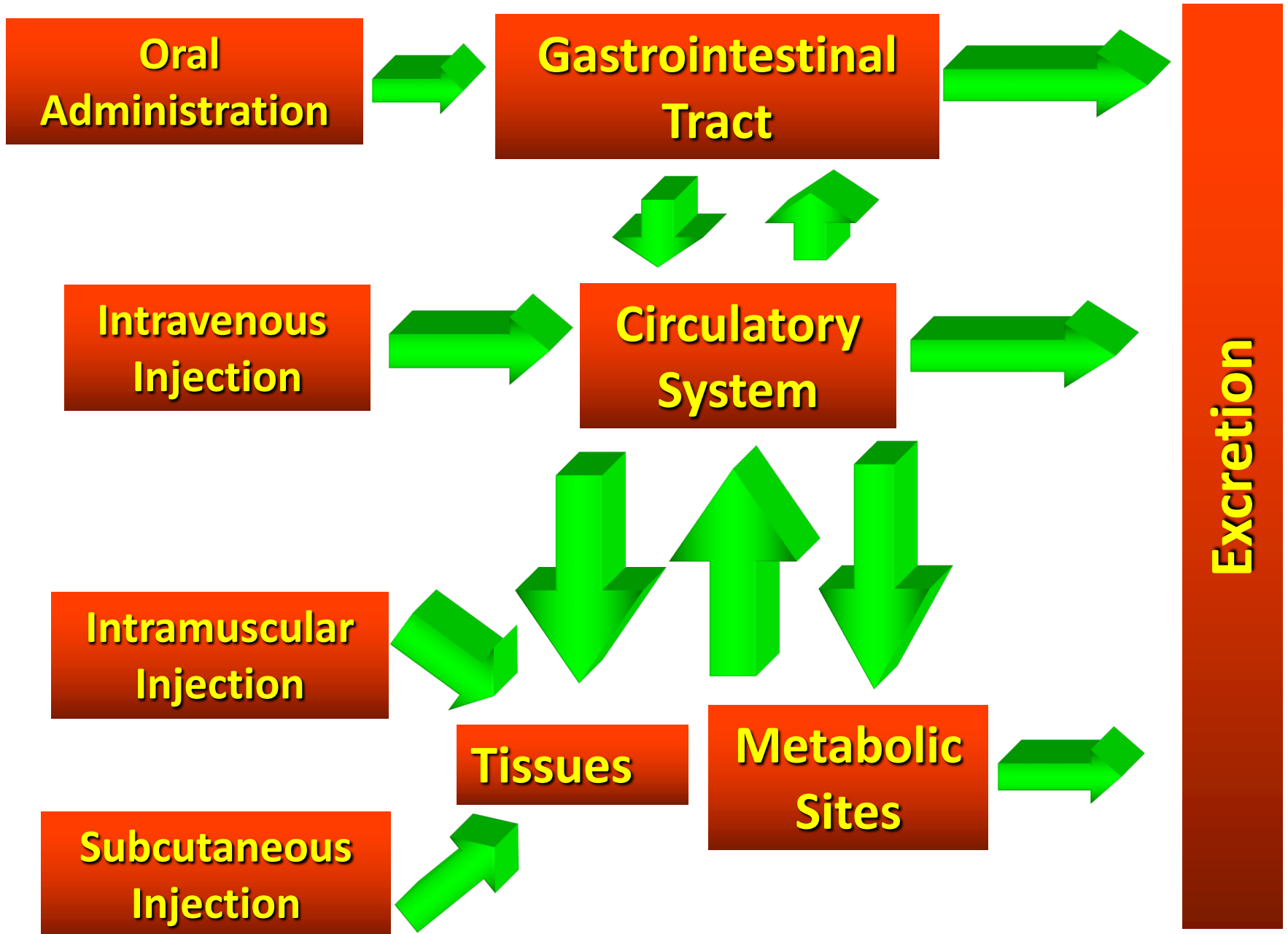


Magnitude of Drug Response

- Depends upon concentration achieved at the site of action
 - Dosage
 - Extent of absorption
 - Distribution to the site
 - Rate/extent of elimination

Pharmacokinetics and Pharmacodynamics





Implications of PK and PD in Drug Delivery

- **The PK and PD of a drug may be affected when administered via different routes**
 - **Examples**
 - **Proteins – oral vs. intramuscular**
- **The PK and PD of a drug delineates its therapeutic window**
 - **Degree of absorption**
 - **Degree of elimination and/or metabolism**
 - **Example**
 - **Tetracycline (infection) – given 6 to 8 hours**
 - **Digoxin (cardiac failure)– given daily**

Factors affecting pharmacokinetics

Solubility

Partition coefficient

Ionization

Several others

Solubility

Drugs must be in solution (not precipitated) to interact with receptors/targets

Drugs have some degree of solubility in both aqueous and lipid compartments (PC)

Solubility is a function of:

- ionization

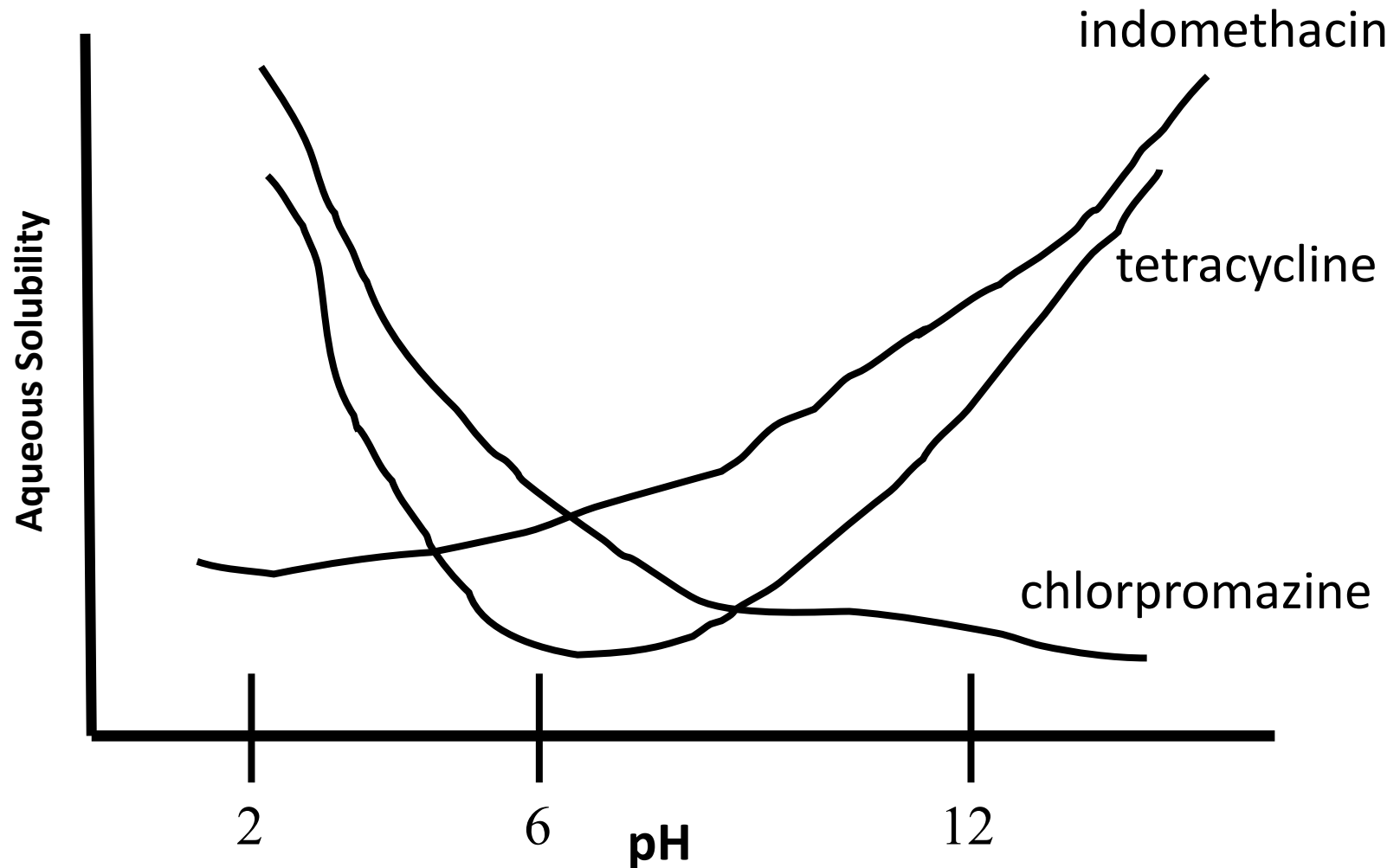
- molecular structure

- molecular weight

- electronic structure

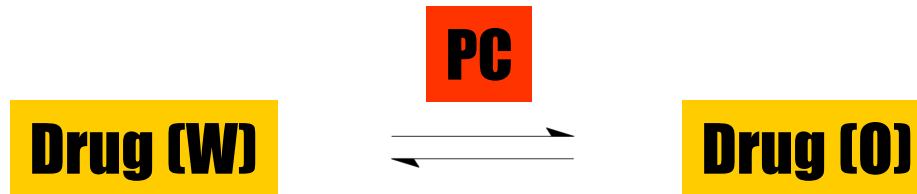
Solubility: Examples

Solubility will be affected by pH



Partition Coefficient

The ratio of the concentrations of a solute in two immiscible or slightly miscible liquids when it is in equilibrium across the interface between them



$$\text{PC} = \frac{[\text{drug}]_o}{[\text{drug}]_w}$$

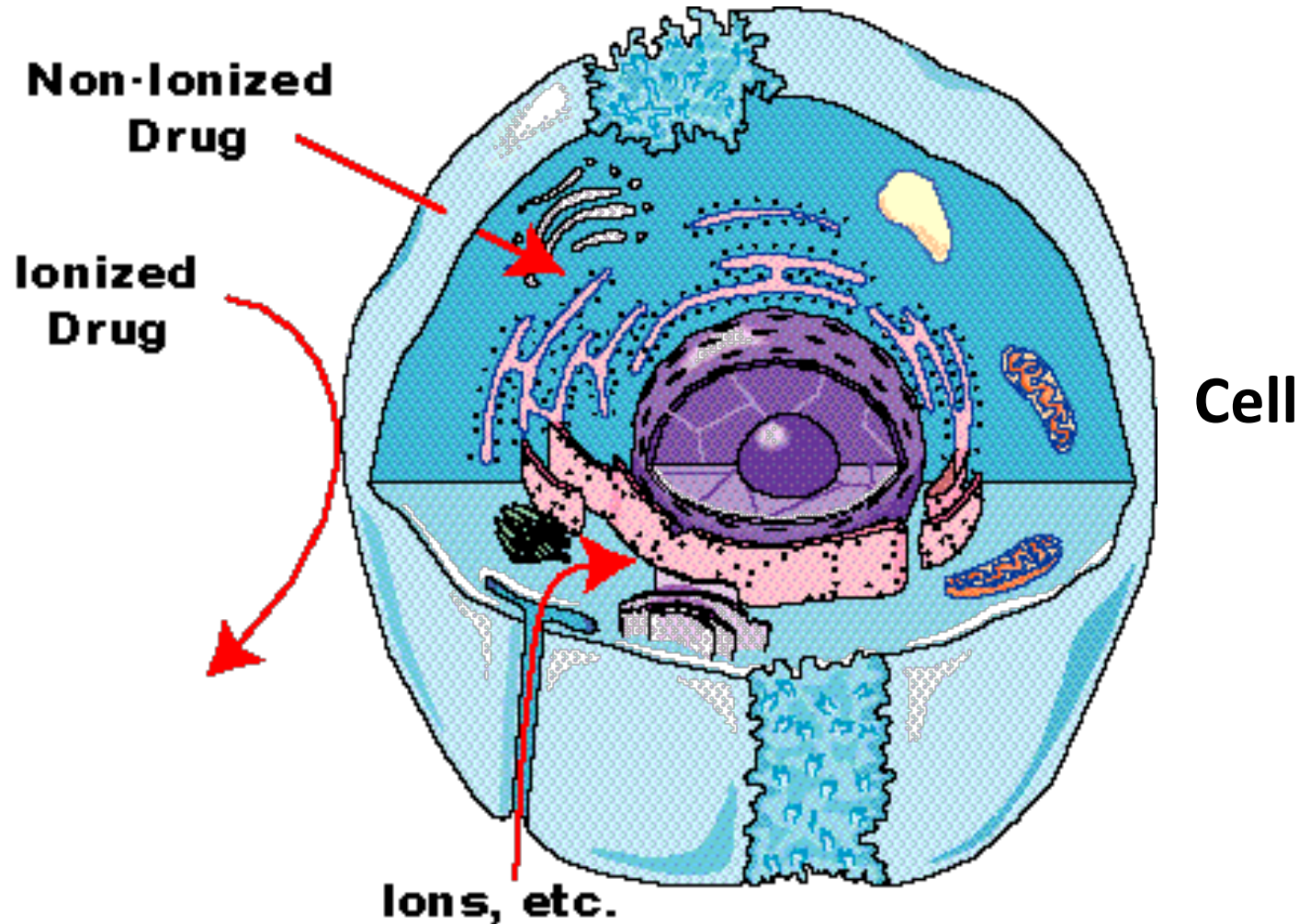
**Thus, PC
describes
the entire
drug.**

Lipid-Water Partition Coefficient

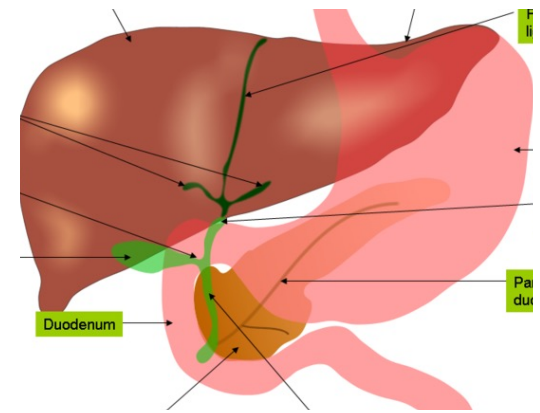
- The higher the lipid/water p.c. the greater the rate of transfer across the membrane
 -  polarity of a drug, by increasing ionization will  the lipid/ water p.c.
 -  polarity of a drug, suppression of ionization will  the lipid/ water p.c.

Drug Ionization

Charged molecules typically do not enter drug by passive diffusion

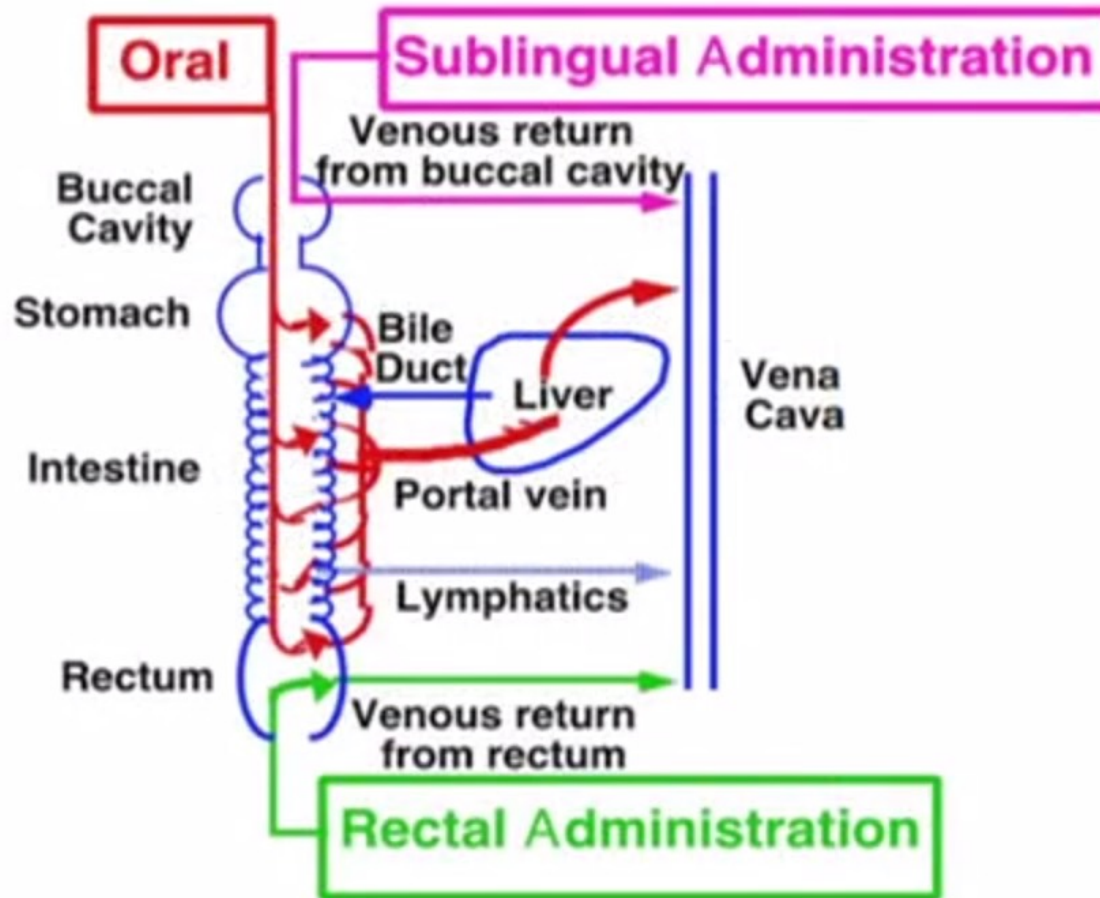


First-pass Effect: Oral Delivery



- Hepatic metabolism of a pharmacological agent
- Absorbed from the gut and delivered to the liver via the portal circulation
- The greater the first-pass effect, the less the agent will reach the systemic circulation
- Applicable when the agent is administered orally

ENTERAL ABSORPTION OF DRUGS



Sublingual route has lower barriers but also has low absorption

Rectal administration prevent first pass metabolism for more than 50% of the drug

Exercise

Let's see how we are doing

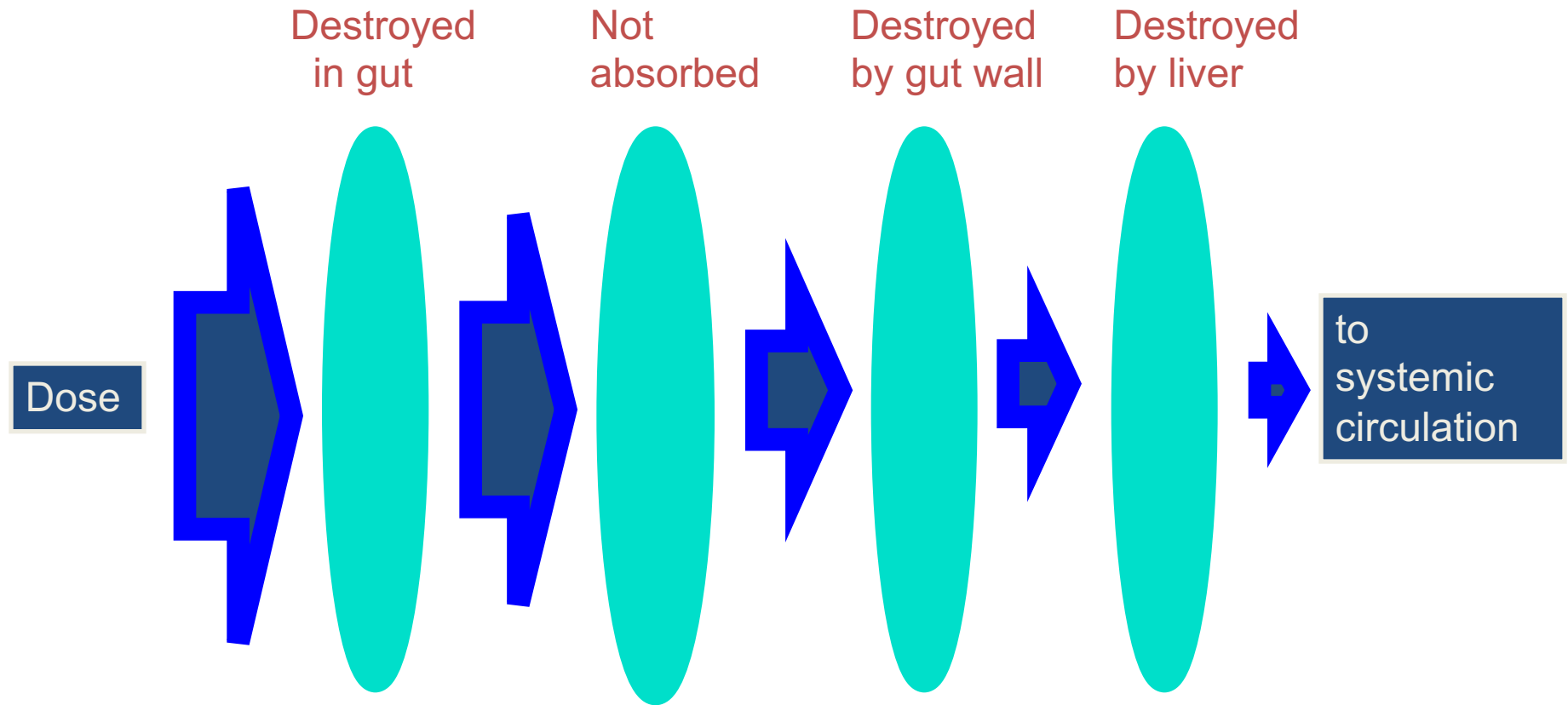
- The first pass effect refers to
 - a. The absorption of a drug from its site of administration
 - b. How the drug reaches its first site of action
 - c. How the amount of a drug can be reduced by metabolism before it reaches the systemic circulation.
 - d. How a drug is metabolized after it reaches the systemic circulation

Answer is c

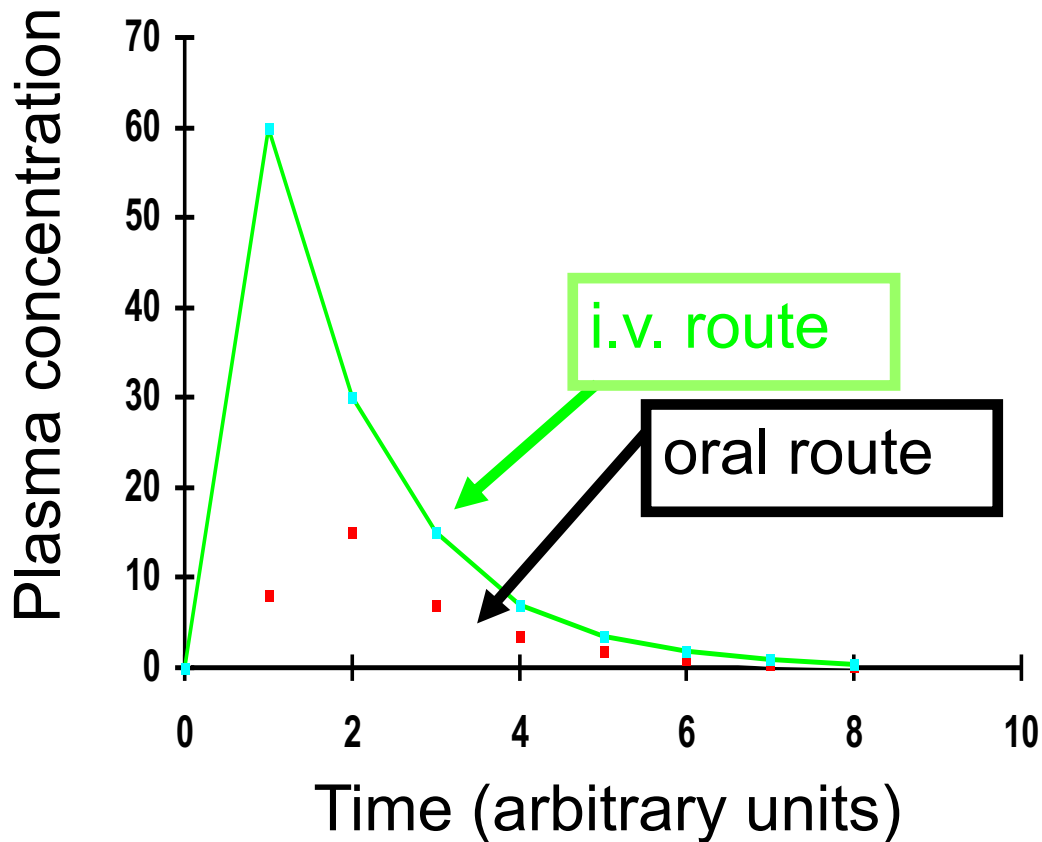
Bioavailability

- Essential to determine what dose will induce the desired therapeutic effect
- Different routes can lead different bioavailability
- 100% of the administered drug is bioavailable when delivered through IV route if it is soluble in aqueous medium
- Other routes may have different efficiencies of drug that are bioavailable even when administered at same dose
- Explains why the same dose may cause a therapeutic effect by one route but a toxic or no effect by another

Bioavailability: Oral Delivery



$$\text{Bioavailability} = \frac{(\text{AUC})_{\text{route}}}{(\text{AUC})_{\text{iv}}}$$



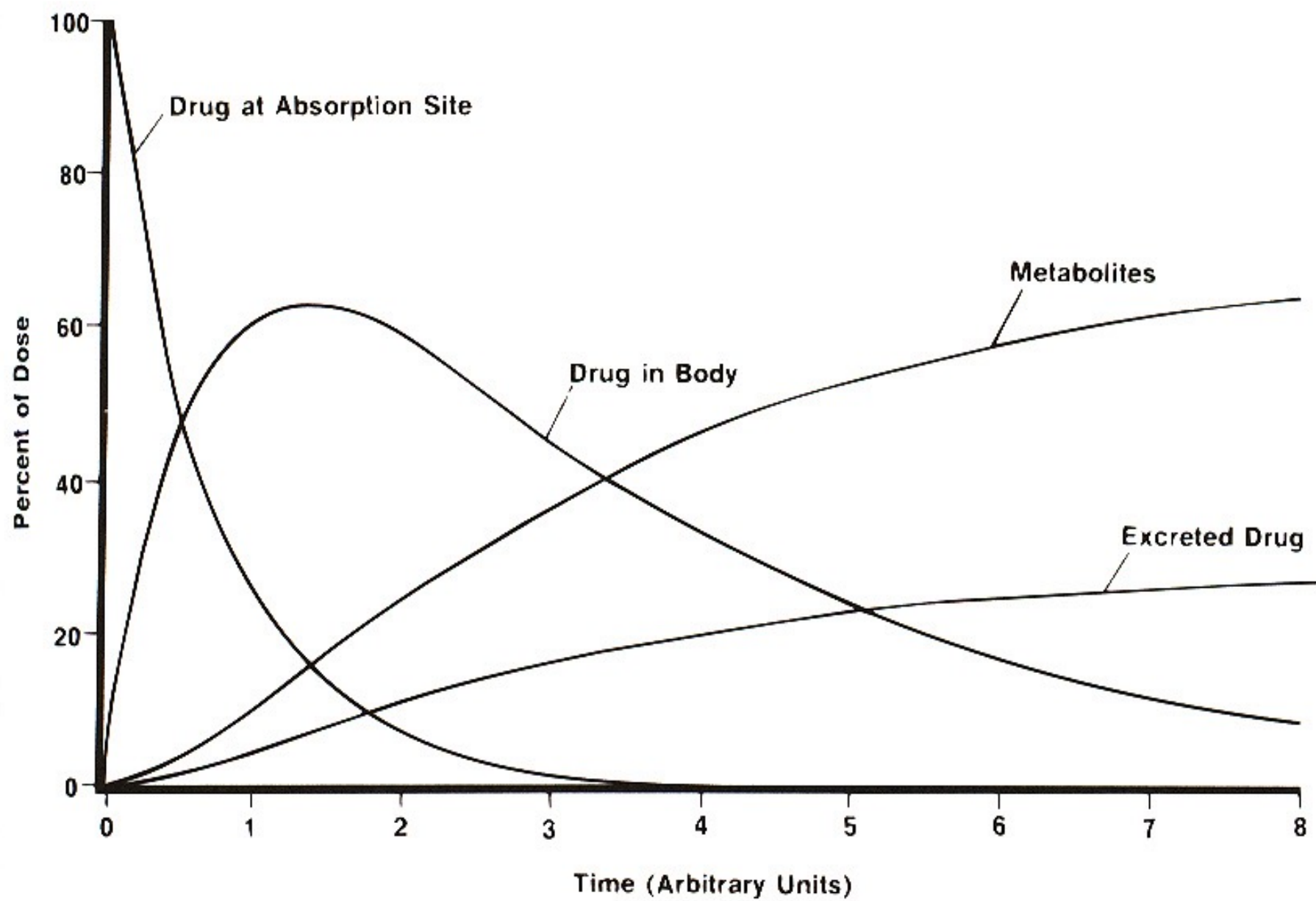


Fig. 3-5. Time course of drug in the body. (From Rowland, M., and Tozer, T.N.: Clinical Pharmacokinetics. 2nd Ed., Philadelphia, Lea & Febiger, 1989.)

Elimination of drugs from the body

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graph TD; A[Elimination of drugs from the body] --> B[KIDNEY<br/>filtration<br/>secretion<br/>(reabsorption)]; A --> C[LIVER<br/>metabolism<br/>secretion]; A --> D[LUNGS<br/>exhalation]; A --> E[OTHERS<br/>mother's milk<br/>sweat, saliva etc.];
```

MAJOR

KIDNEY

filtration
secretion
(reabsorption)

LIVER

metabolism
secretion

MINOR

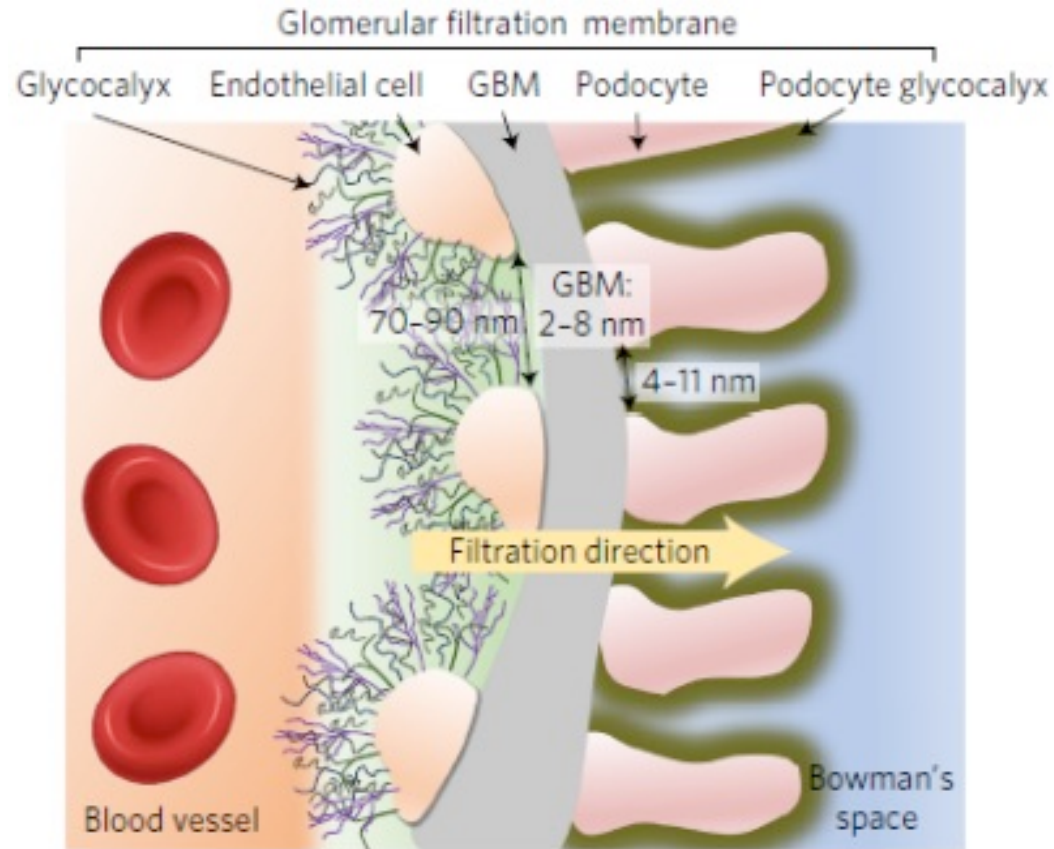
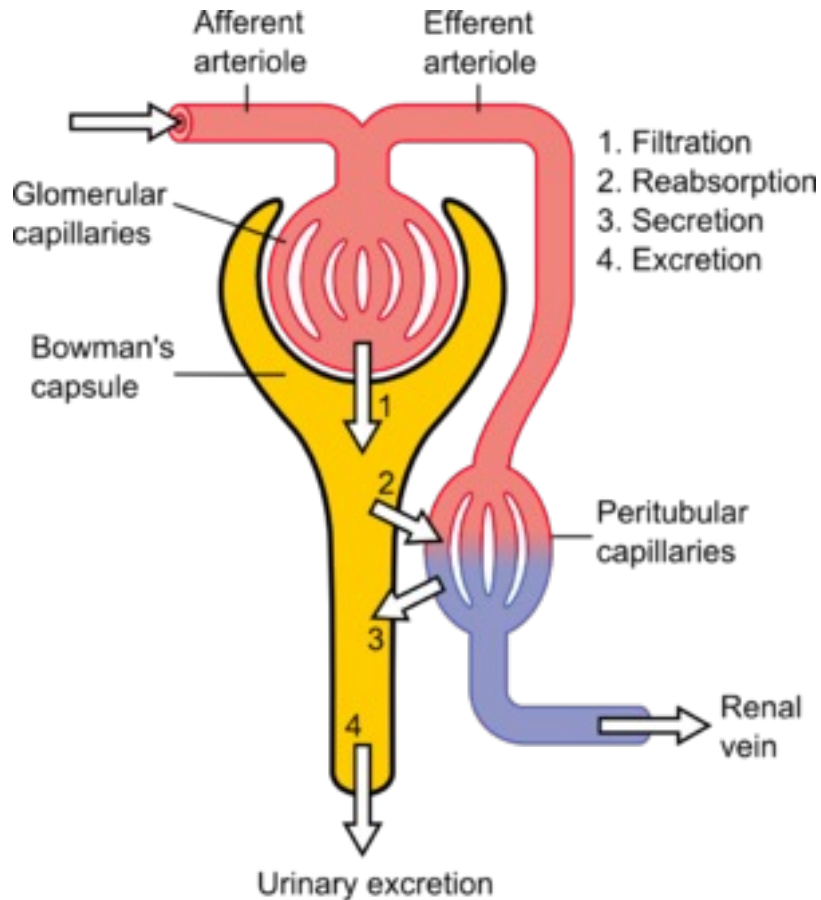
LUNGS

exhalation

OTHERS

mother's milk
sweat, saliva etc.

Elimination by the Kidney



glomerular filtration membrane (GBM)

Particles above 10nm can't be excreted by kidney

Elimination

- **Zero order:** constant rate of elimination irrespective of plasma concentration.
- **First order:** rate of elimination proportional to plasma concentration. Constant *Fraction* of drug eliminated per unit time.

Rate of elimination $\propto$ Amount

Rate of elimination = $K \times$ Amount

Elimination

- **Zero order:** constant rate of elimination irrespective of plasma concentration.
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Rate of elimination $\propto$ Amount

Rate of elimination = $K \times$ Amount

Zero Order Elimination: Ethanol

- ▶ Ethanol is distributed in total body water.
- ▶ Mild intoxication: 1 mg/ml in plasma.
- ▶ How much should be ingested to reach it?

Answer: 50L of water in our body so, 50 g or 63 ml of pure ethanol

130 ml of a strong alcoholic drink like whiskey (2 large peg!)

- ▶ Ethanol has a constant elimination rate = **10 ml/h**
- ▶ To maintain mild intoxication, at what rate must ethanol be taken now?

at 10 ml/h of pure ethanol, or 20 ml/h of drink (less than 1 small peg!!).

DRUNKENNESS → Coma → Death

First Order Elimination

- **Clearance**: volume of plasma cleared of drug per unit time.

Clearance = Rate of elimination $\div$ plasma conc.

- **Half-life of elimination**: time for plasma conc. to decrease by half.

Useful in estimating:

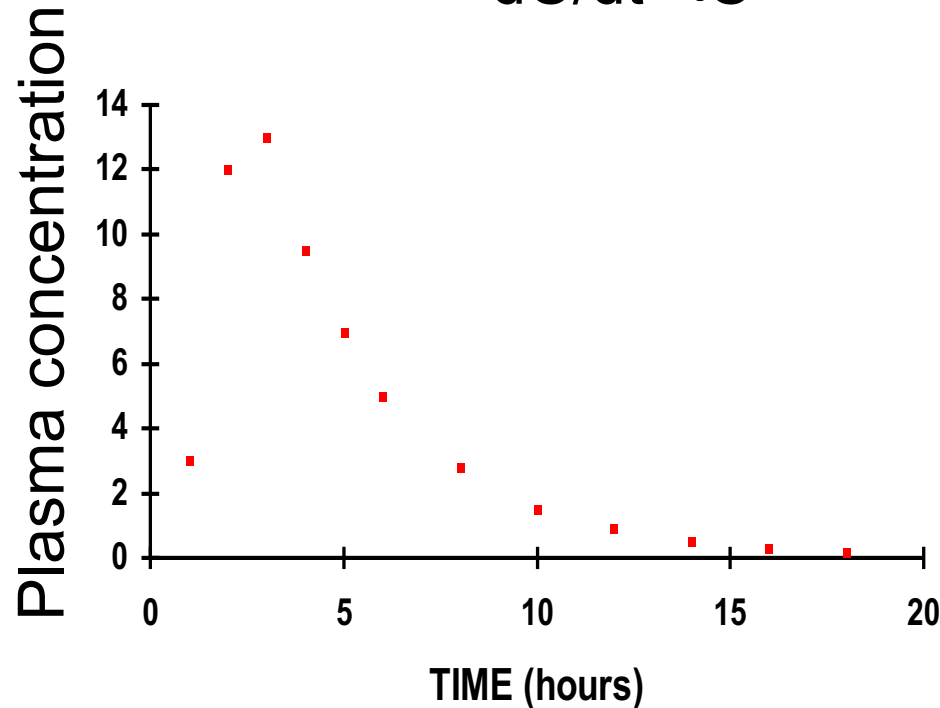
- time to reach steady state concentration.
- time for plasma concentration to fall after dosing is stopped.



$$\log C_t = \log C_0 - \frac{K_{el} \cdot t}{2.303}$$

First Order Elimination

$$dC/dt \propto C$$



$$dC/dt = -k \cdot C$$

$$C_t = C_0 \cdot e^{-K_{el} \cdot t}$$

$$\ln C_t = \ln C_0 - K_{el} \cdot t$$

$$y = b - a \cdot x$$

Biological half-life

Elimination of drugs from the body usually follows first order kinetics with a characteristic half-life ($t_{1/2}$) and fractional rate constant (K_{el}).

An Example – How long will it take for the body to eliminate 100 mg of drug X?

- 100mg of drug in plasma @ time zero and the drug's half life is 4 hours
- 50% of drug is eliminated in one half life
 - 50mg of drug remains in plasma after 4 hours (1 half life)
 - 25mg of drug remains in plasma after 8 hours (2 half lives)
 - 12.5mg of drug remains in plasma after 12 hours (3 half lives)
 - 6.25mg of drug remains in plasma after 16 hours (4 half lives)
- Thus for drugs that are eliminated via first order kinetics >90% of a drug would be eliminated in 4 half lives

Pharmacokinetics: Single compartment model



Entire body act as a single, uniform compartment

Concentration in the plasma and tissues is considered to be uniform

Drug considered to be distributed in single compartment

Pharmacokinetics: Two compartment model

Body divided into central and peripheral compartment

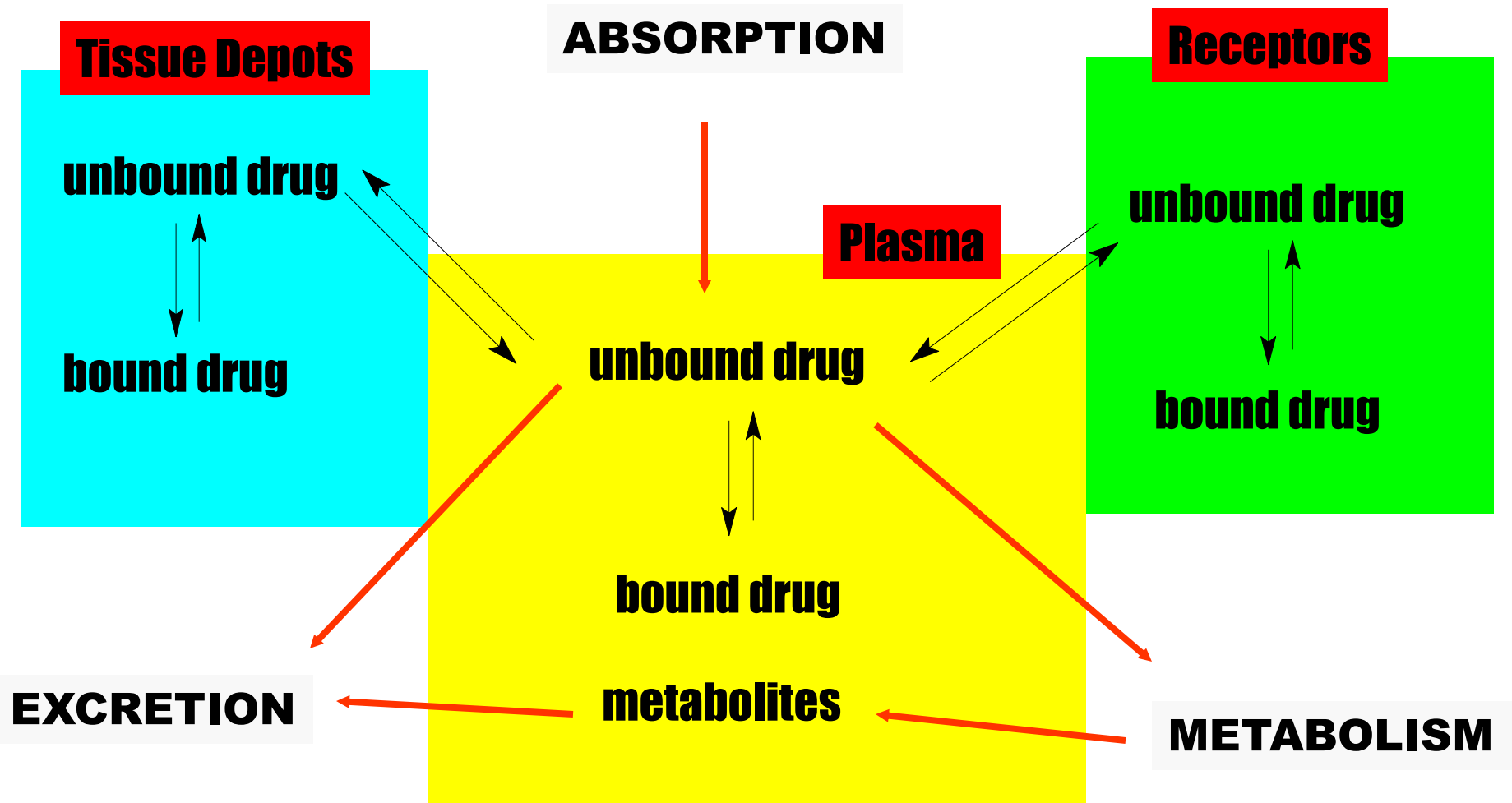
Central compartment: Blood and highly perfused tissues (Kidney, heart, brain etc.)

Peripheral compartment: Rest of tissues (adipose tissue, bone, teeth)

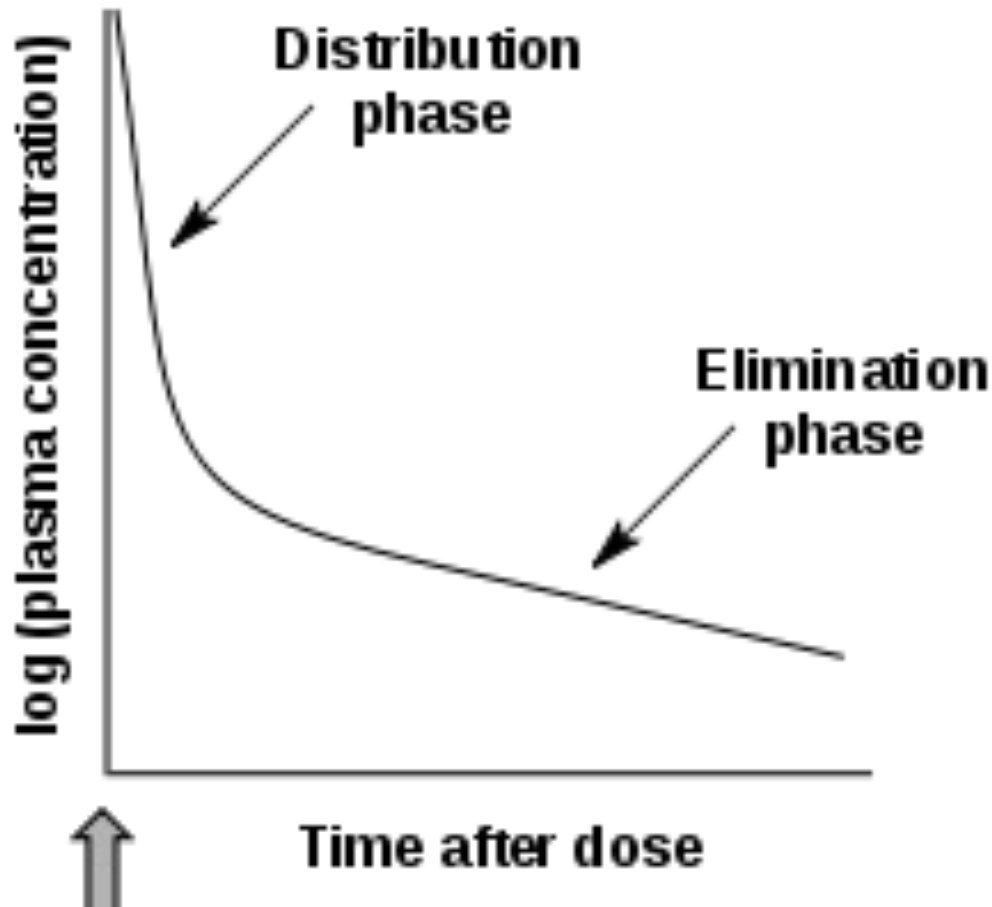
Central and peripheral also have a first order distribution process

Plasma drug concentration declines bi-exponentially

Pharmacokinetics: Two compartment model



Pharmacokinetics: Two compartment model



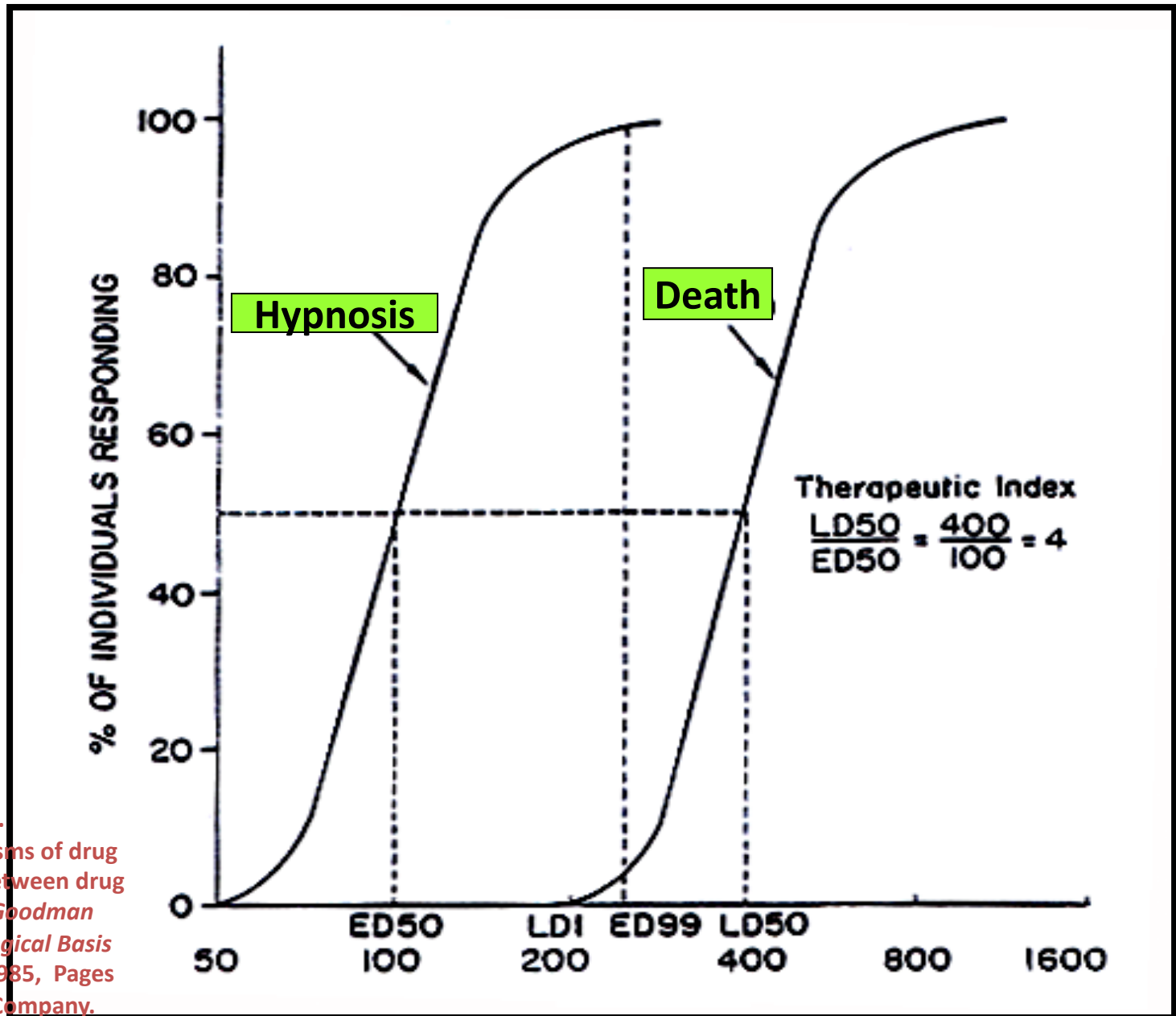
Effectiveness, toxicity, lethality

- ED50 - Median Effective Dose 50; the dose at which 50 percent of the population or sample manifests a given effect
- TD50 - Median Toxic Dose 50 - dose at which 50 percent of the population manifests a given toxic effect
- LD50 - Median Lethal Dose 50 - dose which kills 50 percent of the subjects

Quantification of drug safety

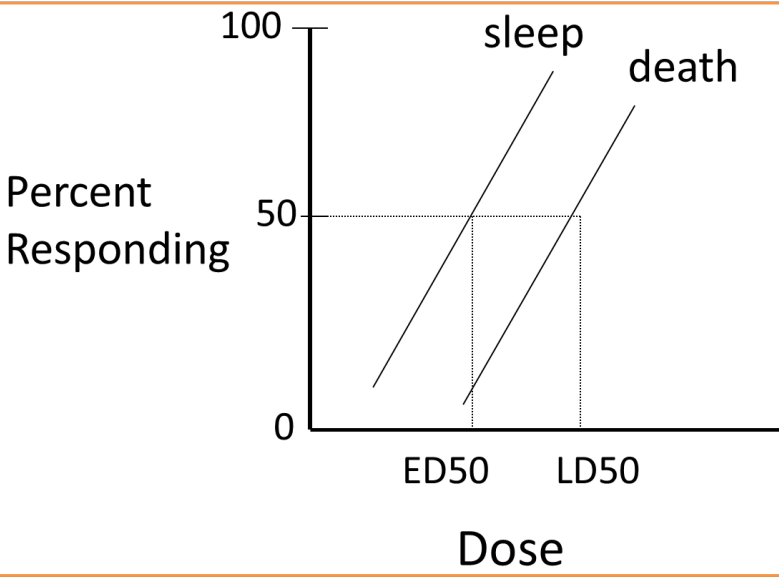
$$\text{Therapeutic Index} = \frac{\text{LD50}}{\text{ED50}}$$

THERAPEUTIC INDEX – AN INDEX OF SAFETY

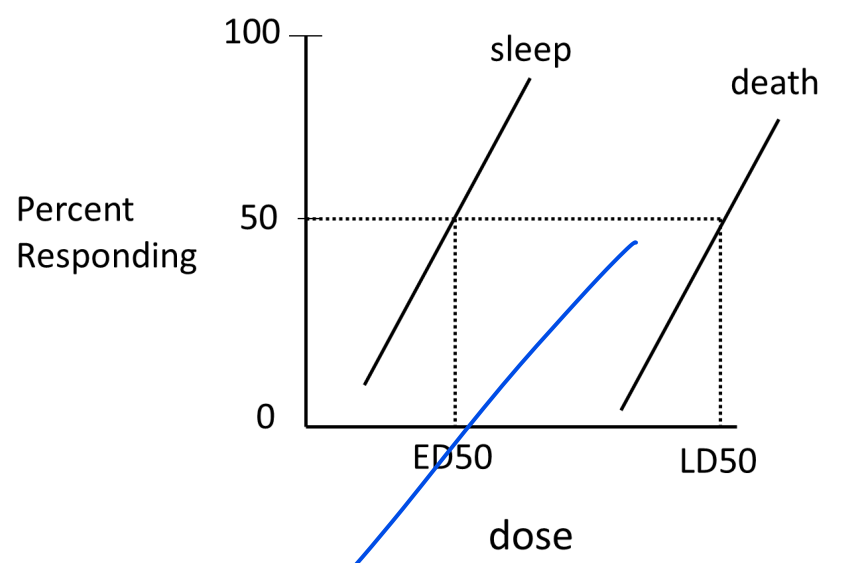


From Ross EM and Gilman AG.
Pharmacodynamics: mechanisms of drug
action and the relationship between drug
concentration and effect.. in *Goodman
and Gilman's The Pharmacological Basis
of Therapeutics*. 7th edition, 1985, Pages
35-48. Macmillan Publishing Company.

Drug A

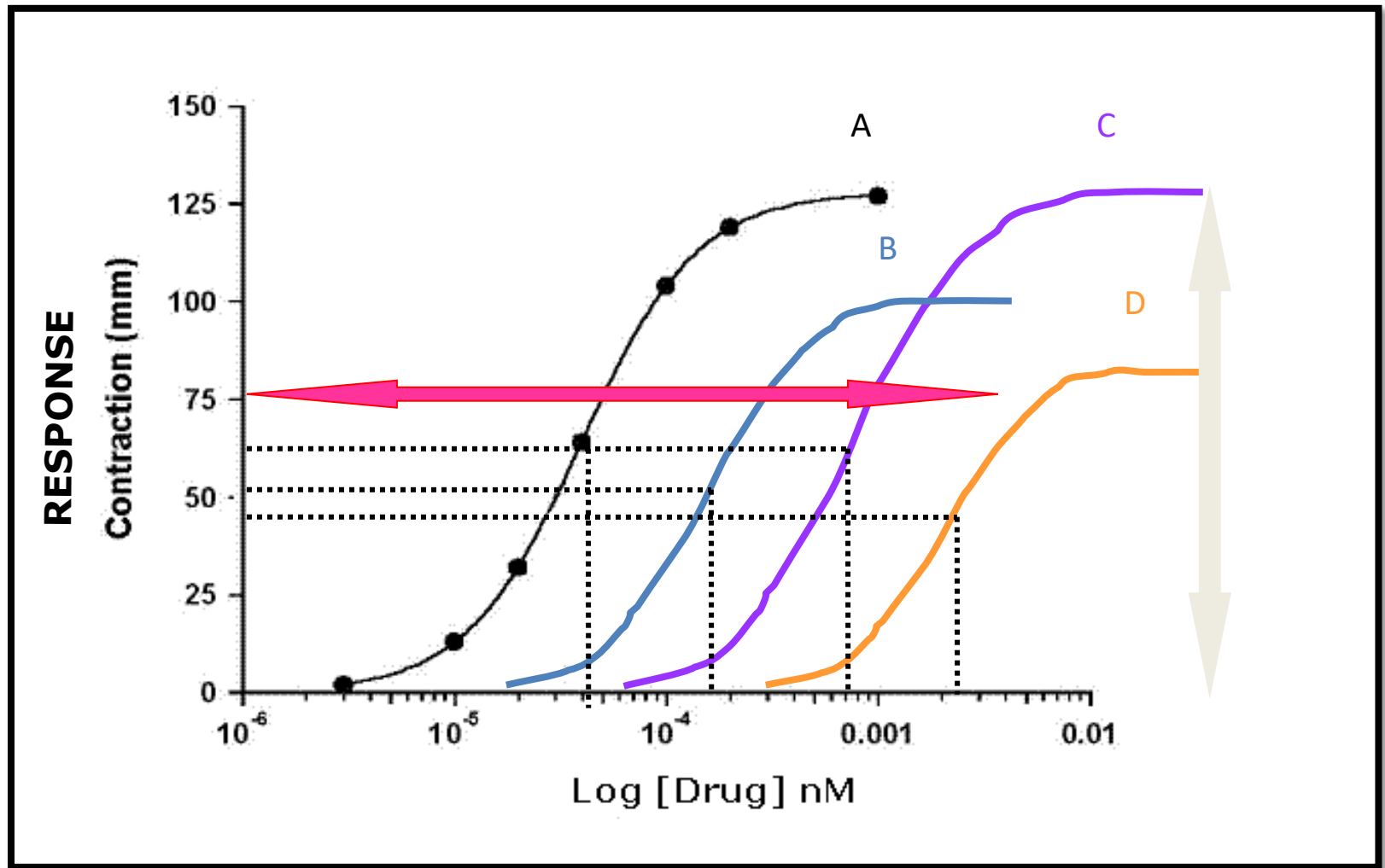


Drug B



Which is better?

Exercise



RANK ORDER OF POTENCY: A > B > C > D

RANK ORDER OF EFFICACY OF RESPONSE: A = C > B > D

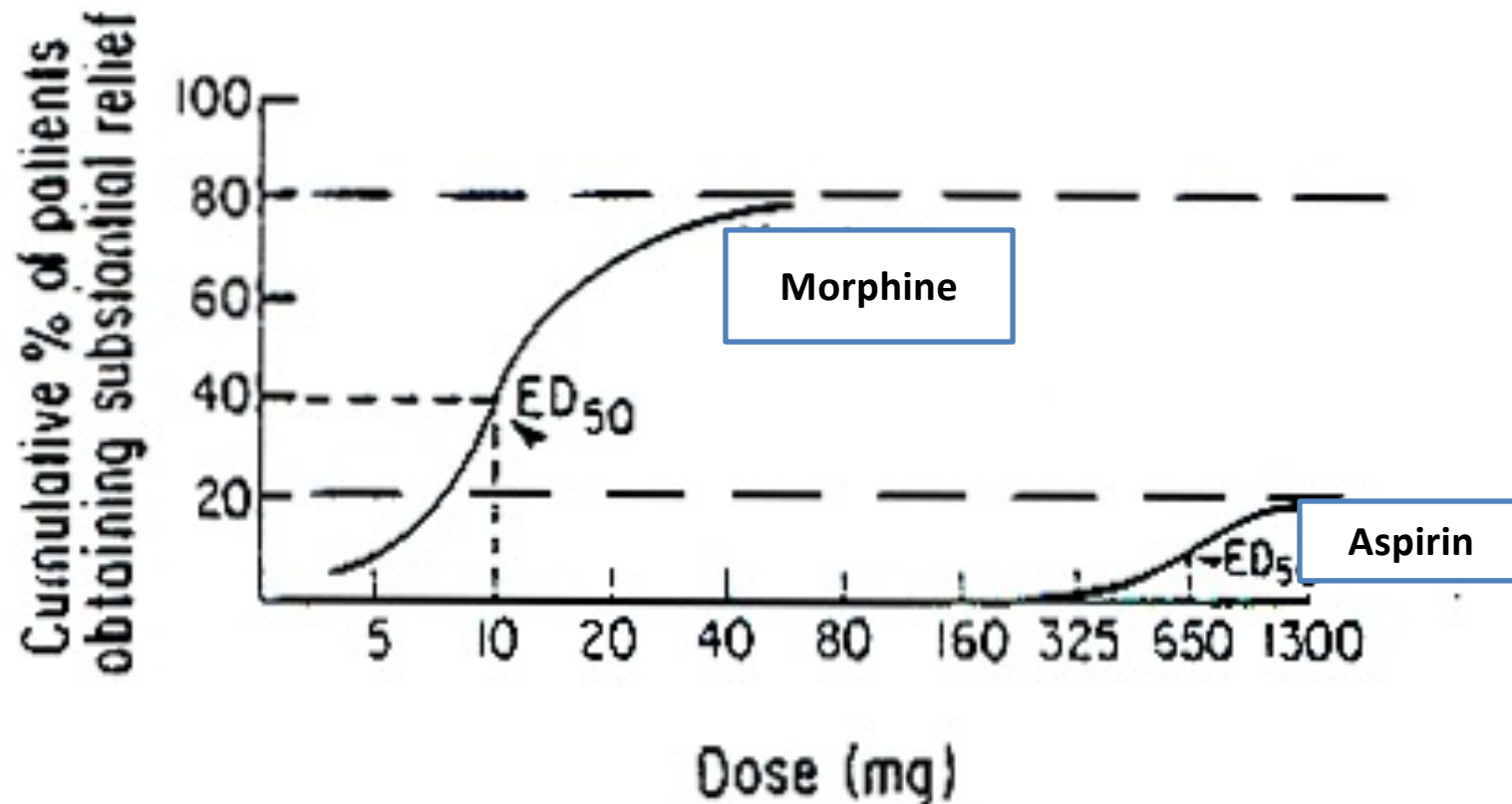
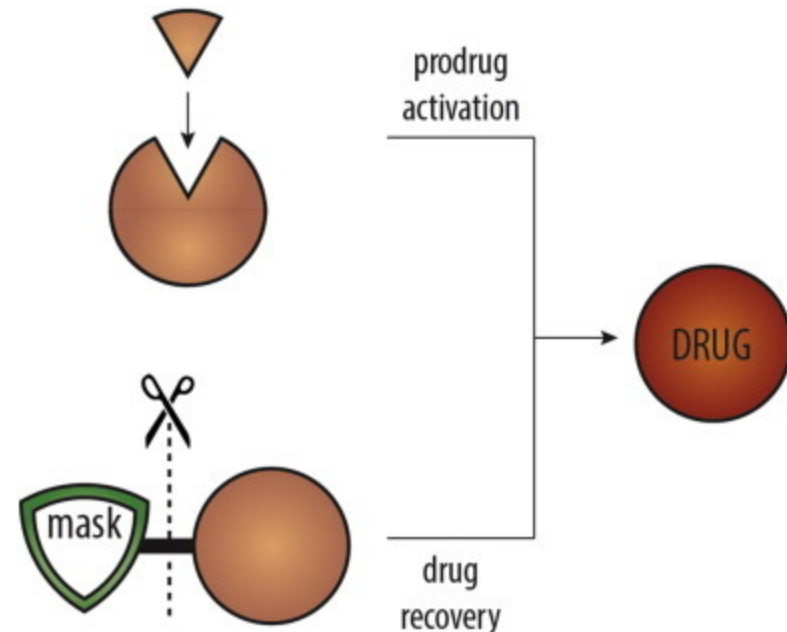


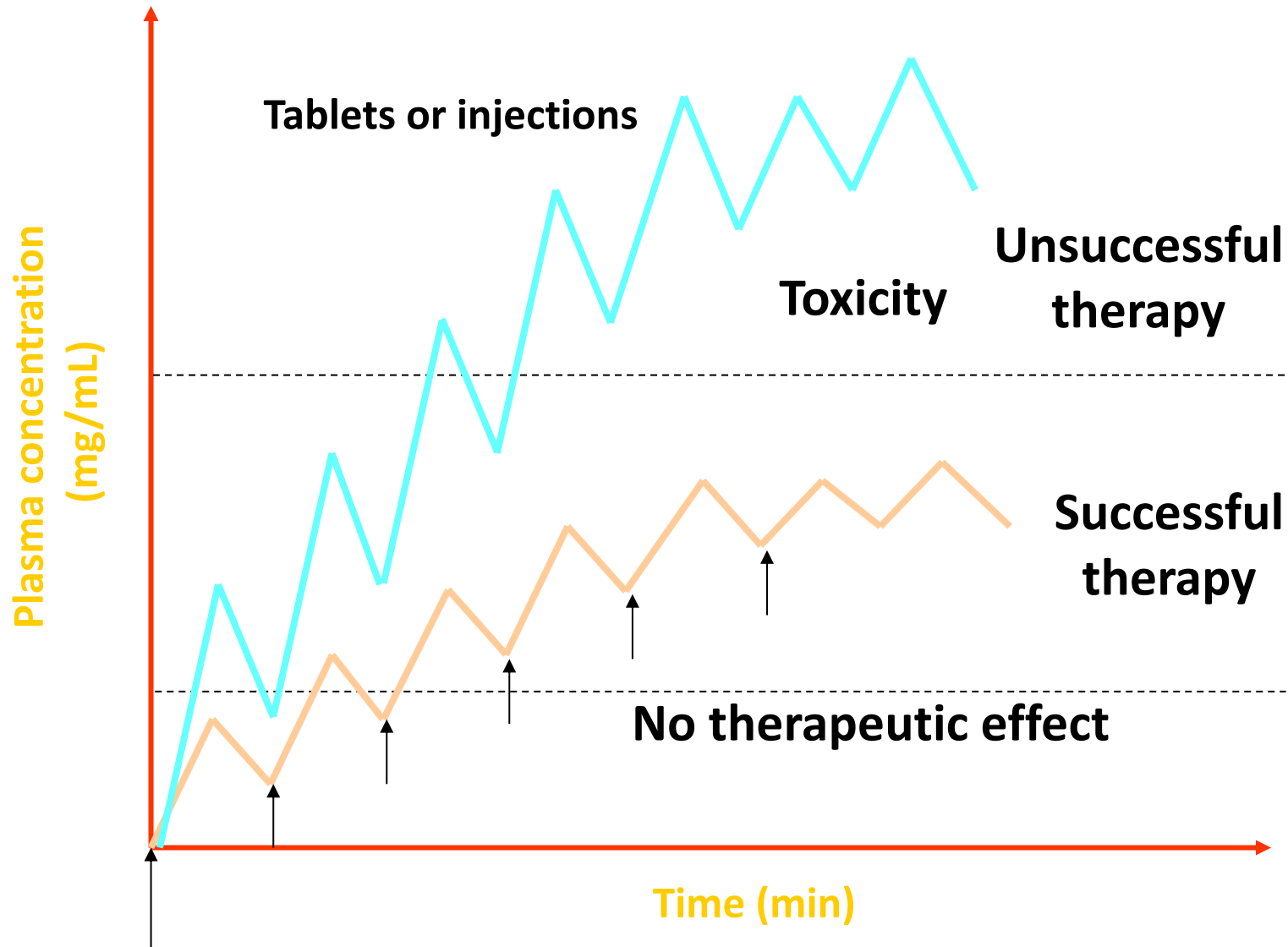
Fig. 1-5. Dose-response curves describing the ability of aspirin and morphine to relieve pain in patients with migraine headache. Note that morphine is both more potent (has a lower ED₅₀), and has a greater overall efficacy (ceiling effect).

Prodrugs

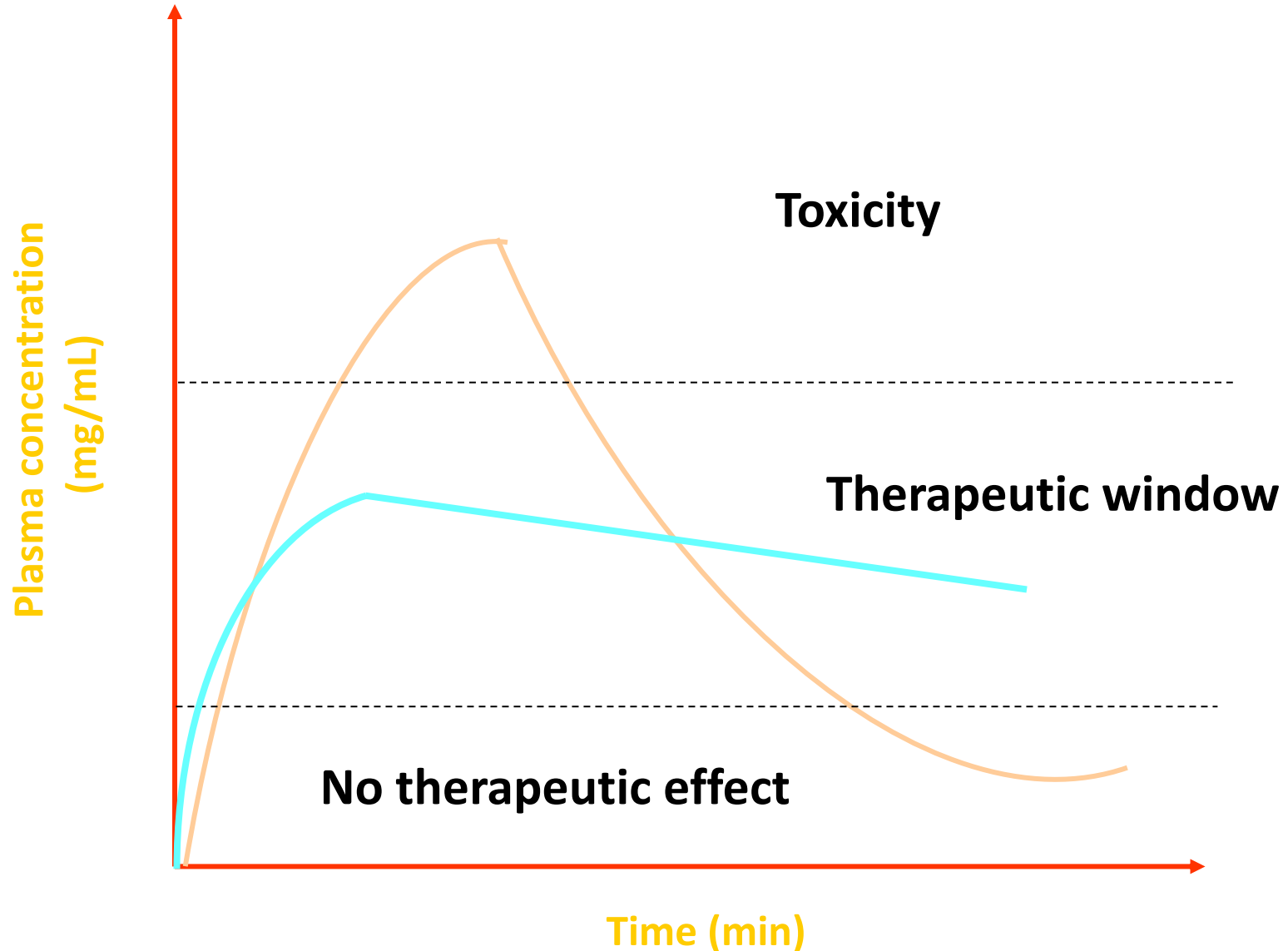
- Compounds that get metabolized into active drug
- Prodrugs by themselves have low or no activity
- May result in increasing bioavailability if the active drug is not well soluble or absorbed
- Improves targeting as the signal (enzymes) to convert the pro-drug to active drug may be localized to target tissue
- 10% of all marketed drugs are prodrugs



Plasma Concentration – Multiple Administrations



Plasma Concentration – Single Administration



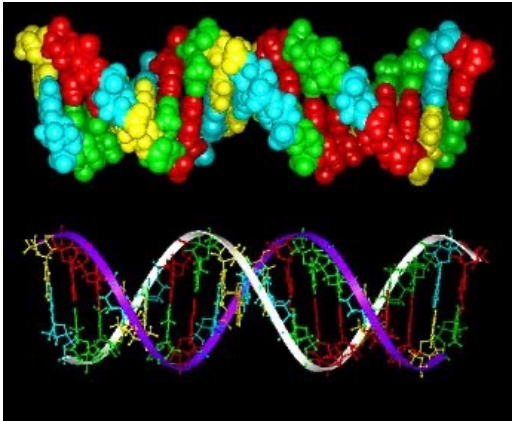
Controlled release and long circulation: Advantages

- **Increase patient compliance**
- **Employ less total drug**
- **Improve efficiency in treatment**

Biomaterials for controlled release

- **External vs Internal**
 - External devices such as pumps, skin patches
 - Lot more flexibility with external material
 - Internal is qualified as “Biomaterial” meaning it is safe and would not cause side effects
- **Biomaterials**
 - Any material that is considered safe for human use
 - Could be synthetic (e.g. synthetic polymer) or natural (e.g. alginate)
 - Could be metallic (e.g. gold, silver), ceramic, polymeric etc.

Polymers – Everywhere in Daily Life



DNA



Starch



Cellulose



Rubber



HDPE



Teflon



Nylon



PVC

Introduction to Polymers

- **What are polymers?**

- Polymers are organic (carbon-based) compounds where each molecule is a long-chain of covalently linked molecular units
- Think of it as a beaded necklace. Each bead is an unit that is repeated to form the necklace (polymer). There can be beads of one kind or of several different kinds.
- E.g. plastics, proteins, nucleic acids, carbohydrates

- **What is a monomer?**

- The basic chemical unit of a polymer, is called a monomer (beads in the necklace)
- E.g. vinyl chloride, ethylene glycol, lactic acid, amino acids, glucose, etc.

Some more terms

- **Macromolecules**

- Refers to very large molecules. Since polymers are generally composed of thousands of monomers, each polymer molecule is very large. Thus polymers are also termed as macromolecules

- **Oligomers**

- These are compounds having a few (unlike “many” in case of polymers) monomers joined together by covalent bonds (usually less than 50 repeating units). E.g. Oligonucleotides, peptides etc.

Types of Polymers: Chemical Structure

A-A-A-A-A-A-A-A-A-A-A-A-A-A

HOMOPOLYMERS

A-B-A-B-B-B-B-A-A-B-B-A-A-A-B

RANDOM COPOLYMERS

A-B-A-B-A-B-A-B-A-B-A-B-A-B-A

ALTERNATING COPOLYMERS

A-A-A-A-A-A-A-B-B-B-B-B-B-B-B

BLOCK COPOLYMERS

A-A-A-A-A-A-A-A-A-A-A-A-A-A-
|
B-B-B-B-B-B

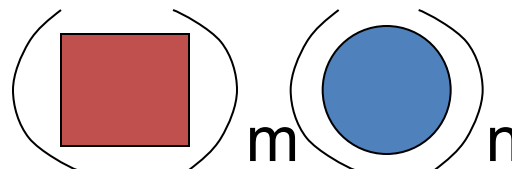
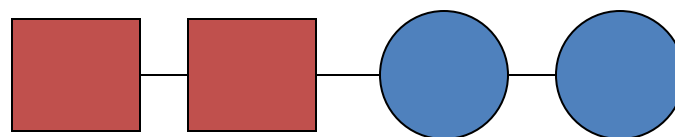
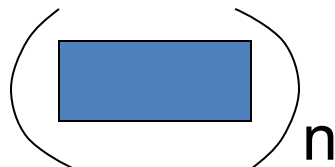
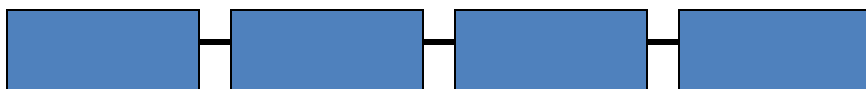
GRAFT COPOLYMERS

Polymer Nomenclature

Polymer types (classes) are named after the functional group linkage in their backbone i.e the bonds between the monomers:
E.g. Polyesters, Polyethers, Polyamides, Polyanhydrides etc.

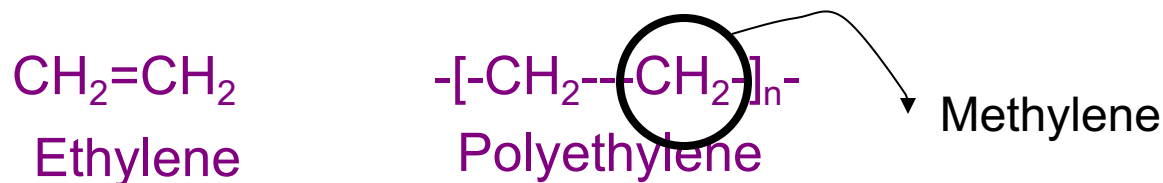
If more than one type of linkage is present then both figure in the names, e.g. polyetherimide

The language (script):



Polymer Nomenclature

- Most common naming process:
 - Poly(monomer), or
 - poly(monomer1-co-monomer2),
 - poly(monomer1-block-monomer2),
 - poly(monomer1-graft-monomer2) etc. i.e. source-based nomenclature
- IUPAC (International Union of Pure and Applied Chemistry) nomenclature: rarely used in common literature, only as references
- Example of a source-based polymer name:



IAPUC name of this polymer will be poly(methylene)

How are polymers synthesized?

Simply speaking: Individual monomer molecules (same or different kinds) are covalently linked together by a variety of chemical reactions to produce the polymer molecule.

- Carother's classification
 - **Addition polymerization**: Each repeat unit has the same number of atoms as the monomers
 - **Condensation polymerization**: Repeat units have less number of atoms than the monomers because of formation of small molecular weight reaction products (e.g. H_2O , HCl etc.)
- Modern classification: Based on how the reaction occurs
 - **Step polymerization**
 - random union of monomer molecules
 - **Chain polymerization**
 - polymer chain grows only by reaction of a monomer with a reactive end-group on the growing chain

Molecular weight (molar mass)

Molar mass of a polymer = Mass of one mole of the polymer (i.e. 6.023×10^{23} molecules of the polymer). Usually expressed in units of g/mol or daltons. Molecular weight of 100,000 daltons means molar mass of 100,000 g/mol i.e. one mole weighs 100,000 grams

For a homopolymer: molar mass (M) = xM_o , where M_o is the molar mass of a repeat unit (monomer) and x = average degree of polymerization (i.e. the number of repeat units in a polymer chain)

The polymerization process inherently creates polymer molecules of different lengths. Therefore **we always measure “AVERAGE” molecular weights and “AVERAGE” degree of polymerization for polymers.**

Average molecular weights

Number average molecular weight (M_n):

Defined as $M_n = \sum X_i M_i$, where X_i = mole fraction of molecules having a molar mass of M_i

If N_i = Number of moles with mass M_i , then

$$M_n = \sum N_i M_i / \sum N_i \dots\dots\dots(1)$$

NOTE: This is the arithmetic mean of the molar mass distribution

$N_i M_i$ is also same as w_i , the weight of molecules having a molar mass of M_i i.e. $M_n = \sum w_i / \sum (w_i / M_i)$

Example 1: Polymer sample consisting of 9 moles with molecular weight 30,000 and 5 moles with molecular weight 50,000.

$$\begin{aligned} M_n &= \{[9 \text{ mol} \times 30000 \text{ g/mol}] + [5 \text{ mol} \times 50000 \text{ g/mol}]\} / (9+5) \text{ mol} \\ &= 37,000 \text{ g/mol} \end{aligned}$$

Example 2: Sample with 9 grams of molecular weight 30,000 and 5 grams with molecular weight 50,000

$$M_n = (9 + 5) \text{ g} / \{[9 \text{ g} / 30,000 \text{ g/mol}] + [5 \text{ g} / 50000 \text{ g/mol}]\} = 35,000 \text{ g/mol}$$

Average molecular weights

Weight average molecular weight (M_w):

Defined as $M_w = \sum W_i M_i$, where W_i = weight fraction of molecules having a molar mass of M_i

If N_i = Number of moles with mass M_i , then

$$W_i = N_i M_i / \sum N_i M_i = w_i / \sum w_i \dots\dots\dots(2)$$

$$\text{Therefore, } M_w = \sum N_i M_i^2 / \sum N_i M_i = \sum w_i M_i / \sum w_i \dots\dots\dots(3)$$

[Second order average]

Example 1: Polymer sample consisting of 9 moles with molecular weight 30,000 and 5 moles with molecular weight 50,000.

$$\begin{aligned} M_w &= \{9 \text{ mol } (30000 \text{ g/mol})^2 + 5 \text{ mol } (50000 \text{ g/mol})^2\} \text{ divided by} \\ &\quad \{9 \text{ mol } (30,000 \text{ g/mol}) + 5 \text{ mol } (50,000 \text{ g/mol})\} \\ &= 40,000 \text{ g/mol} \end{aligned}$$

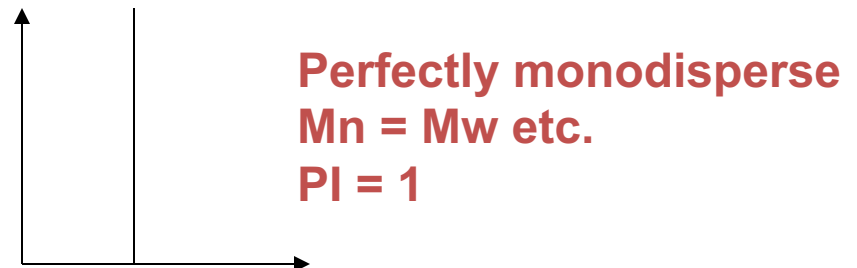
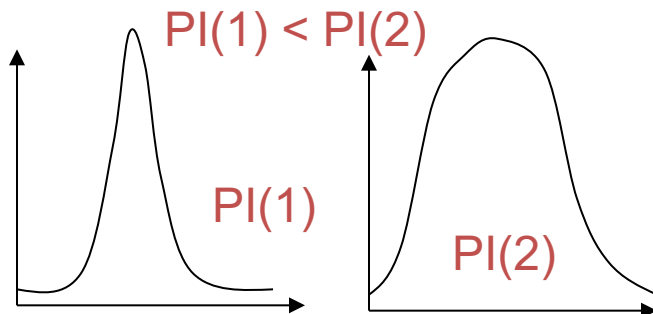
Example 2: Sample with 9 grams of molecular weight 30,000 and 5 grams with molecular weight 50,000

$$M_w = \{9\text{g } (30,000 \text{ g/mol}) + 5\text{g } (50000 \text{ g/mol})\} / (9 + 5) \text{ g} = 37,000 \text{ g/mol}$$

NOTE: M_w IS ALWAYS GREATER THAN or EQUAL TO M_n

Polydispersity index (PI)

- Polydispersity index is a measure of the width of molecular weight distribution
- It is defined as the ratio between the weight and number average molecular weights (M_w / M_n)
- Therefore, by definition **PI is ≥ 1 , since $M_w \geq M_n$**
- For a perfectly monodisperse polymer $PI = 1$
- Greater the PI, broader is the molecular weight distribution

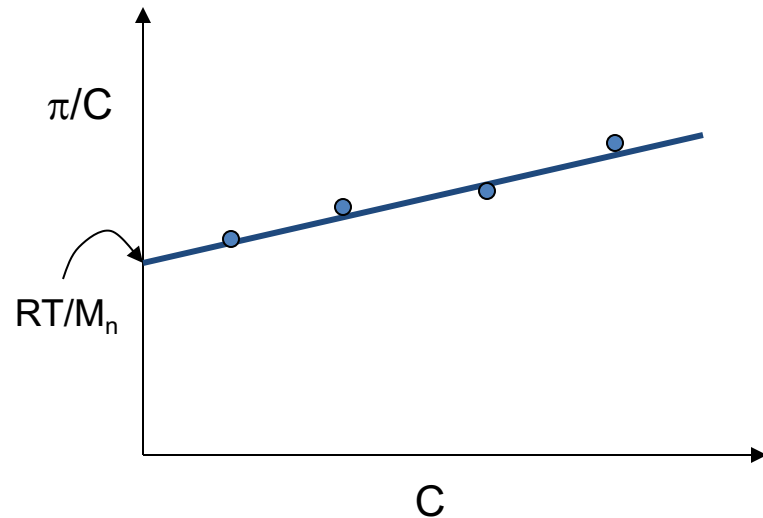
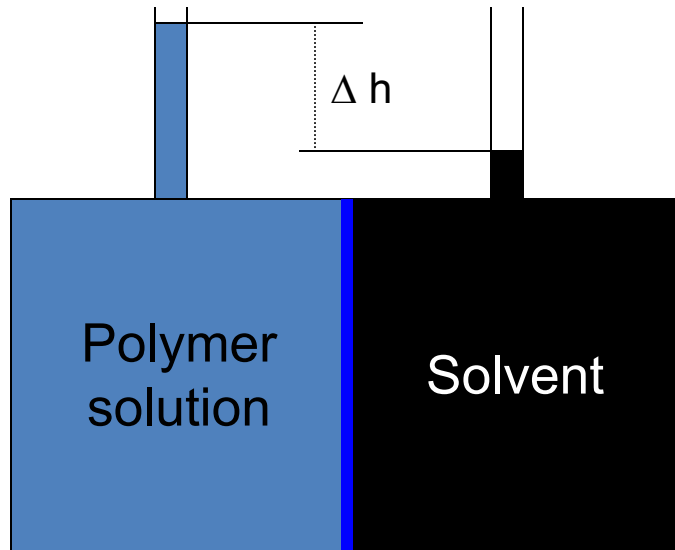


Molecular weight measurements

- **Number average molecular weight** (Calculated with techniques that measure “number” of particles – colligative properties)
 - End group analysis
 - Membrane osmometry
 - Vapor pressure osmometry
 - Cryoscopy
 - Refractive index measurements
- **Weight average molecular weight**
 - Light scattering
 - Ultra-centrifugation
- **Viscosity measurements**
- **Molecular weight distribution:**
 - Gel permeation Chromatography
 - Mass spectrometry
 - Fractional solution
 - Thin layer chromatography

Membrane osmometry

- Widely used method for measuring M_n (measuring colligative property)
- Ideal for polymers with M_n between 50,000 and 2 million



Osmotic pressure is related to molecular weight by the van't Hoff equation

$$(\pi/C) = (RT/M_n) + A_2C \dots\dots\dots(1)$$

Where π , the osmotic pressure is given by $\pi = \rho g \Delta h \dots(2)$

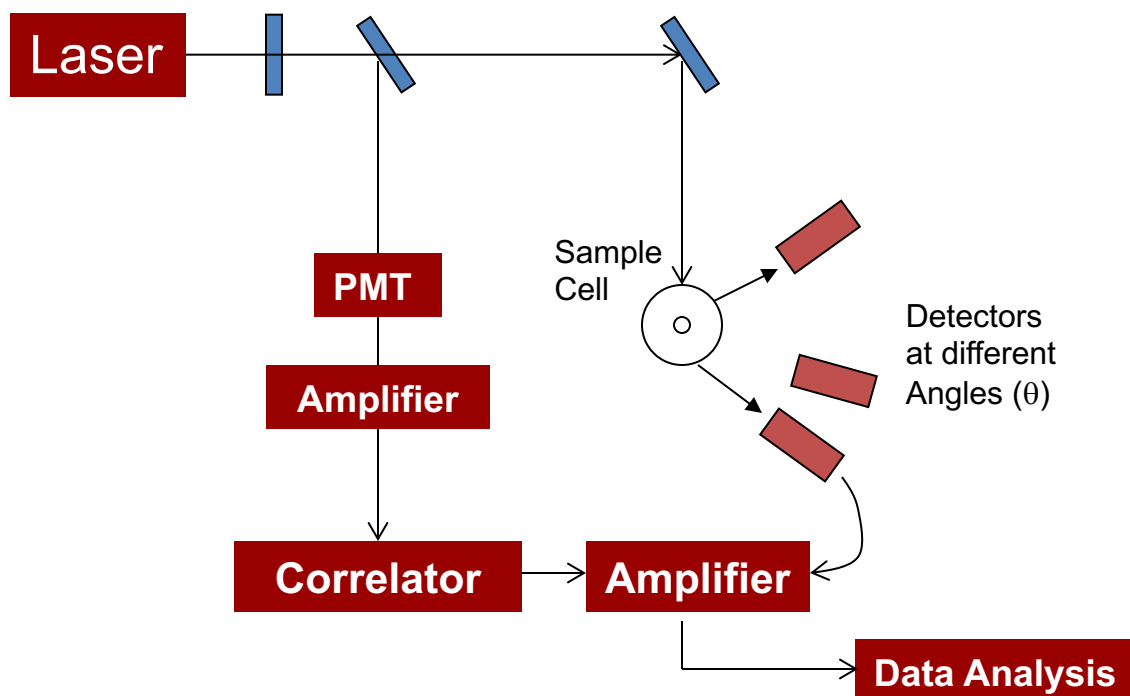
R = Gas constant, T = Temp ($^{\circ}\text{K}$), C = conc. in g/L, ρ = solvent density (g/cc)

g = acceleration due to gravity (m/s^2) and Δh = as shown in figure

At $C = 0$ (extrapolation), the intercept provides M_n

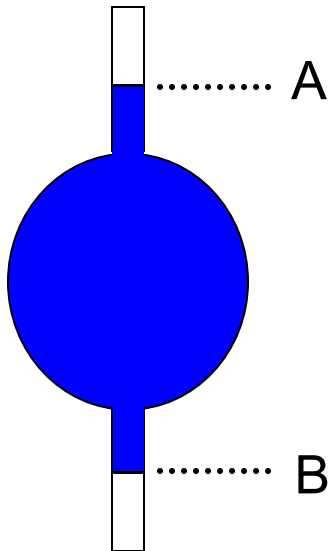
Light scattering: Setup

- Widely used for molecular weight determination
- Gives Mw
- Also used to measure hydrodynamic radius
- Useful between molecular weights of 10,000 to 10 million



Viscosity Measurements

- Widely used method to calculate **molar mass M_n (viscosity average)**
- Lot of manufacturer's will only specify **intrinsic viscosity** values in their CA, instead of molecular weights
- **Higher the molar mass, higher is the viscosity**



Efflux time: Time for the polymer solution to travel from A to B (measured with a stop watch)

More time → more viscous → higher molecular weight

Viscosity Measurements

Efflux time of pure solvent (no polymer) = t_o

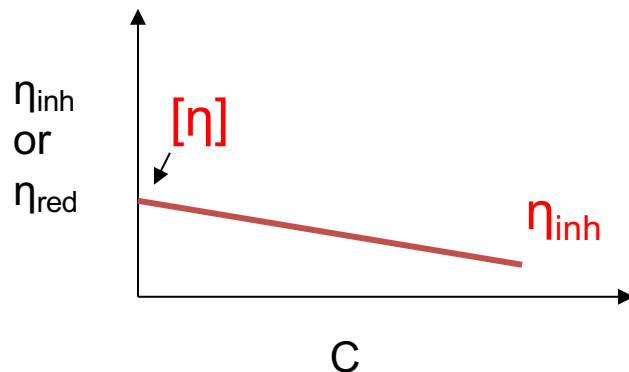
Efflux time of polymer solution at concentration “c” = t_c

Relative viscosity = t_c / t_o (always greater than 1) = η_r

Inherent viscosity = $\ln(\eta_r) / c = \eta_{inh}$

Intrinsic viscosity $[\eta]$ is defined as

$$[\eta] = \lim_{c \rightarrow 0} (\eta_{inh})$$



Molecular weight and viscosity is related by the Mark-Houwink-Sakurada equation: $[\eta] = KM^a$

K and a are experimentally determined from $\log [\eta]$ vs $\log [M]$ plots

Gel permeation chromatography (GPC) or Size exclusion chromatography (SEC)

- Extremely powerful method for determining molecular weight distribution of a polymer.
- Specialized columns packed with beads of a porous gel with porosity in the range of $50 - 10^6 \text{ \AA}^3$

Morphological properties:

Crystallinity

- When a polymer is slowly cooled from the melt state, it can form ordered structures (similar to crystals). Such polymers are called crystalline (semi-crystalline)
- Polymers with regular, compact structures and strong intermolecular forces, such as hydrogen bonds have high degrees of crystallinity
 - as crystallinity increases, the polymer becomes more opaque due to scattering of light by the crystalline regions
- **Melt transition temperature, T_m** (or Crystalline melting point): The temperature at which all crystalline regions melt i.e. crystallinity disappears
 - as the degree of crystallinity increases, T_m increases

Morphological properties: Crystallinity

Crystallinity has major effects on

a. Mechanical properties

Increased stiffness

Less flexible

b. Diffusion rates

Less permeable to diffusing molecules

c. Rates of hydrolysis

Low penetration of water and hence slow hydrolysis

Morphological properties:

Amorphousness

- Polymers that doesn't form an ordered structure when cooled from a melt are called amorphous polymers
- Amorphous polymers are also referred to as glassy polymers
 - They lack crystalline domains that scatter light and are **transparent**
 - On heating, amorphous polymers are transformed from a hard glass (stiff, brittle) to a soft, flexible, rubbery state
- **Glass transition temperature, T_g :**
 - The temperature at which a polymer undergoes a transition from a hard glassy state (amorphous solid) to a rubbery state

$$T_g < T_m$$

Glass Transition Temperature (T_g)

- **Motion of polymer chains** possible above T_g , while below T_g only atomic or small molecule motions occur
- Number of **polymer properties e.g. heat capacity, density, permeability, dielectric constant etc. changes** abruptly at T_g
- Polymers are **brittle below T_g and flexible (rubbery) above T_g . Rubber ball shatters in liquid nitrogen.**
- **Depending on usage, proper polymer is chosen**
 - Silicones have an extremely low T_g and T_m , they are always fluid at relevant temperatures (breast implants)
 - Polystyrene, PMMA is in a glassy state at room temp and above ($T_g \sim 100^\circ\text{C}$). So they can be made into hard plastic objects
- Chain mobility is critical in diffusion. **Diffusion below T_g is much lower than diffusion above T_g**

Morphology: Effect of cooling rate

- **Poly(ethylene terephthalate)**, abbreviated PET can be made with crystalline domains ranging from 0% to 55% depending on how it is processed. Completely amorphous PET is formed by **cooling the melt quickly**
 - plastic beverage bottles are PET: What do u think they are?
 - with a low degree of crystallinity
- By **cooling slowly**, more molecular diffusion occurs, chains become more ordered and crystalline domains form
 - PET with a high degree of crystallinity can be drawn into textile fibers and tire cords (Dacron)

Factors affecting T_g

- **Molecular weight:** End groups have higher mobility (higher free volume). Low molecular weight polymers have higher number of ends per unit volume and hence lower T_g . Higher molecular weight polymers have higher entanglement, less mobility and hence higher T_g .
- **Bulky side groups** hinders chain motion making chains stiffer. Hence higher T_g (PMA vs. PMMA)
- **Strong intermolecular attractions** → higher T_g

Higher T_m → Higher T_g

Can be measured using Differential Scanning Calorimetry

Biomedical Polymers

- **Biopolymers:** Polymers that can be safely used in biological or medical applications or polymers that are naturally present in organisms are called biopolymers
- **Synthetic biopolymers**
 - Chemically synthesized polymers (does not occur in nature)
 - Designed specifically for particular usage, including bio-molecule delivery, implants, tissue engineering, wound dressing, prosthetics etc.
- **Natural biopolymers**
 - Occurs naturally (in plants and other organisms)
 - Isolated and purified for biomedical usage including drug delivery, tissue engineering, wound applications etc.

Advances in Biomaterials for Drug Delivery

Owen S. Fenton, Katy N. Olafson, Padmini S. Pillai, Michael J. Mitchell,
and Robert Langer**

Advances in biomaterials for drug delivery are enabling significant progress in biology and medicine. Multidisciplinary collaborations between physical scientists, engineers, biologists, and clinicians generate innovative strategies and materials to treat a range of diseases. Specifically, recent advances include major breakthroughs in materials for cancer immunotherapy, autoimmune diseases, and genome editing. Here, strategies for the design and implementation of biomaterials for drug delivery are reviewed. A brief history of the

lose their pharmacological activity due to changes in environmental factors such as moisture, temperature, and pH, which can occur in the body or during storage. As the biotechnology industry continues to develop new classes of biopharmaceuticals, improved fundamental understanding of how drug delivery affects safety and efficacy, along with new delivery

Synthetic vs. natural biopolymers

Synthetic polymers	Natural polymers
Chemically synthesized from monomers <i>E.g: PLGA, PET, PEG etc.</i>	Isolated as polymers from plants or organisms <i>E.g.: Cellulose, chitosan, hyaluronic acid, chondroitin sulfate, gelatin, collagen, proteins and DNA etc.</i>
Can be tailored according to desired properties	In its native form, the properties cannot be controlled
Can be modified or functionalized according to applications	Modification is often difficult, but some polymers can be modified to alter properties for specific usage (semi-synthetic)
Large scale production is easily feasible	Large scale production can be often difficult

Natural polymers in biomedical applications

Table 1. A summary of the main properties and applications of polymeric biomaterials

Polymer	Main applications and comments
Natural polymers	
Proteins and protein-based polymers	Absorbable, biocompatible, nontoxic, naturally available, typically elastic materials used as implants and in tissue engineering.
Collagen	Absorbable sutures, sponge wound dressing, drug delivery microspheres.
Albumin	Used in cell and drug microencapsulation.
Poly(amino acids)	Usually poly(α ,L-amino acids), examples include poly(α ,L-lysine), poly(α ,L-glutamic acid), poly(aspartic acid) etc. Advantages: nontoxic, nonantigenic and biocompatible. Used as oligomeric drug carriers.
Polysaccharides and derivatives	
<i>From vegetable sources</i>	
Carboxymethyl cellulose	Cell immobilization via a combination of ionotropic gelation and polyelectrolyte complex formation (e.g. with chitosan), in drug-delivery systems and dialysis membranes.
Cellulose sulphate	Component of polyelectrolyte complexes for immunoisolation. Complex-forming ability is highly sensitive to degree of acylation.
Agarose	Largely used as supporting materials in clinical analysis and as an immobilization matrix.
Alginate (marine sources, algae)	Excellent gel-formation properties; its relative biocompatibility; microstructure and viscosity are dependent on the chemical composition (batch-to-batch variations). Used as immobilization matrices for cells and enzymes, controlled release of bioactive substances, injectable microcapsules for treating neurodegenerative and hormone-deficiency diseases.
Carrageenan	Excellent thermoreversible properties. Used for microencapsulation.
<i>From human and animal sources</i>	
Hyaluronic acid	Excellent lubricant, potential therapeutic agent.
Heparin and heparin-like glycosaminoglycans	Antithrombotic and anticoagulant properties. Extensively used in surgery. Some are candidates for ionotropic gelation and capsule formation.
<i>Microbial polysaccharides</i>	
Dextran and its derivatives	Excellent rheological properties. Plasma expander. Widely used as a drug carrier.
Chitosan and its derivatives	Biocompatible, nontoxic, excellent gel- and film-forming ability, natural polycation. Widely used in controlled-delivery systems (e.g. gels, membranes, microspheres).

Desirable properties of biopolymers

- Non-immunogenic (Low inflammatory response)
- Non toxic (does not cause cell or tissue death / damage)
- Properties (mechanical, chemical, electrical) can be tailored for specific applications
- Properties are conducive of easy scale-up (i.e. mass production)

In some cases (especially in most drug delivery applications) it is desirable that the polymer does not remain in the body for a prolonged period. It should degrade and adsorbed, excreted or metabolized under physiological conditions.

Note: Such physiologic degradation or erosion is NOT desirable for a large number of biomedical applications (implants, prosthetics etc.)

Choosing biomedical polymers

- Application.... Application..... Application?
- Route of administration
 - Intravenous, parenteral, cutaneous, mucosal
- Bio-compatibility
 - What kind of cells and tissues will be exposed, host interactions
- Is bio-erosion important or even required?
 - Permanent or temporary material
- Surface properties and compositions
 - Is protein adhesion necessary or deleterious, interactions with other cells, specific or non-specific targeting

Choosing biomedical polymers (cont.)

- Mechanical properties
 - How much load does the device need to bear, how long?
 - Does one need to produce arbitrary shapes and designs
- Environmentally sensitive
 - Stimuli sensitive properties, swelling or drug release?
- Permeability
 - Solute diffusion, porosity

Box 1. Polymer properties needed for specific biomaterial applications

Dental applications

Properties and design requirements

Stability and corrosion resistance in the aggressive oral environment; plasticity and easy processing; strength and fatigue resistance; coating activity; good adhesion and integration to the surrounding tissues; low allergenicity; long lifetime.

Applications and polymers used to date

Poly(methylmethacrylate)-based resins as filling materials; prosthesis. Mechanical properties improved by copolymerization. Poly(zinc acrylates); polyamides.

Future needs

New materials with improved elasticity and bone-soft-tissue integration.

Ocular

Properties and design requirements

Composition and design: solutions, ointments, gels and films. Adhesion, gel- or film forming ability, solubility or hydrophilicity. Biocompatibility and site-specific bioadhesive properties.

Applications and polymers used to date

Hydroxypropylmethylcellulose and poly(vinyl alcohol) for artificial tears; hyaluronic acid; polyacrylamide gels.

Future needs

Polymeric biomaterials with 'intelligent' surfaces to improve bioadhesion.

Orthopaedic

Properties and design requirements

Strength and resistance to mechanical constraints and fatigue; stress modelling; surface designed for cell-specific interactions; sterilizability; good bio- and histocompatibility; durability (for long-term bone implants); good integration with bone and muscles; easily fabricated; injectable or *in situ* moulding, depending on use.

Applications and polymers used to date

Bone cements: degradable scaffold for bone and cartilage replacement; cell and growth-factor delivery. Poly(methylmethacrylate); poly(lactic acid), poly(glycolic acid) and their copolymers.

Future needs

New synthetic biodegradable scaffolds with improved biological and mechanical properties.

Vascular devices

Properties and design requirements

Wear-and-tear resistance; fatigue resistance; lubricity; lack of thrombus or embolism formation and chronic inflammatory response; biostability; sterilizability; porosity (tissue ingrowth); mechanical integrity.

Applications and polymers used to date

Catheters, vascular graft materials, heart valves. Silicones, Teflon®, poly(urethanes).

Future needs

Interdisciplinary approach to create biomimetic materials with enhanced cell interaction; bioresponsive polymers; computer-aided design of micro- and ultrastructures of the implants.

Biortificial organs and soft-tissue implants

Properties and design requirements

The same mechanical properties as the surrounding tissues; no chronic response to toxic leachables; integration with natural tissues without fibrous reaction; porosity; creep resistance.

Applications and polymers used to date

Artificial skin and breast implants (collagen and silicone based). Artificial liver (hepatocyte transplantation) – poly(lactic acid) and poly(glycolic acid).

Future needs

Creation of complex, three-dimensional structures that can be used like artificial organs. Development of a total artificial heart.

Encapsulation

Properties and design requirements

Permeability and diffusive properties to allow optimum transport through the membrane; membrane thickness and mechanical stability must remain intact in physiological environments.

Applications and polymers used to date

Hydrogel- and polyelectrolyte-complex-based microcapsules for encapsulating cells, enzymes, hormones, proteins for bioartificial organs and disease treatments. Alginate, chitosan, synthetic polyelectrolytes.

Future needs

New polymers and strategies for encapsulation can result in the preparation of more-effective encapsulation systems; responsive polymers, novel techniques, non-traumatic implantation procedures to preserve the integrity of the protective membrane.

Drug delivery

Properties and design requirements

Design depends on the route of administration and release site; well-defined structure and functionality.

Applications and polymers used to date

Hydrogels, membranes, networks and copolymers with different structures and applications.

Future needs

Improving the design of the systems, new bioresponsive and 'intelligent' materials.

Biocompatibility and Biodegradability

- Biocompatibility

- Property of materials that determines the extent of adverse physiological reactions (inflammation, immune response, toxicity etc.) when the material is placed inside the body. Lower the adverse reactions, more biocompatible is the material.

- Biodegradability

- In general refers to the ability of materials (polymers) to break down into smaller units (by physiological conditions) that are either adsorbed (metabolized) or excreted by the body.
- This is a very generalized term used interchangeably with other similar terms: bio-erosion, bio-resorption etc. We will visit this issue in more details later in this course

Biocompatibility: Host reactions to polymers

- Mostly a result of normal physiological processes that are designed by nature to act as host defense. The key is to use polymers that the body does not get “mad” about i.e. can tolerate and co-exist
- All materials (polymers) will interact with the host. It is the extent of that interaction that is important
- Some key interactions:
 - Blood-material interactions
 - Protein adsorption, coagulation, platelet adhesion
 - Complement activation
 - Leukocyte adhesion / activation
 - Encapsulation , scar tissue formation
 - Foreign body reaction
 - Infection and inflammation

All these will be covered in the inflammation part of the course

Assessment of Biocompatibility

- **Cyto-toxicity measurements**
 - Morphological cell culture assays
 - Direct contact
 - Elution or extract dilution
 - Biochemical cell function assays
 - Enzymatic activity following polymer exposure
- **In-vivo tissue reactions**
 - Histological evaluation following implantation
 - Time point studies to determine extent of reactions
- **Blood compatibility of polymers**
 - Coagulation studies
 - Hemolysis measurements
 - In-vivo experiments to evaluate systemic toxicity

Improving biocompatibility

- Surface modification with more compatible polymers. E.g. Polyethylene glycol, hyaluronic acid (biomimetic polymers)
- Surface modification to reduce protein adsorption (i.e. reduce blood interactions) – e.g. making the surface more hydrophilic, less cationic etc.
- Concurrent release or delivery of anti-inflammatory agents at the implant site
- Using alternative routes to avoid systemic toxicity (delivery into the blood would have higher chances of systemic toxicity)

Biodegradability

- Most drug delivery devices are by necessity temporary → widest use of biodegradable polymers
- Lets take care of the terminology first (Consensus Conference of The European Society of Biomaterials, 1987)
 - **Biodegradation**: Degradation by biological molecules e.g. enzymatic or microbial degradation.
 - **Bioerosion**: Erosion of a polymer into water soluble products under physiological conditions. This includes both physical and chemical processes (suggested by Heller, 1987)

Physical modes of bioerosion

- **Bulk erosion**

- The rate of water penetration into the solid device exceeds the rate at which the polymer is eroded (i.e. transformed into water soluble products)
 - leads to an erosion process that occurs throughout the entire volume of the solid device (particle or implant)
 - most hydrophilic polymers, devices with high porosities would be bulk eroding (this is the case with most polymers currently used in drug delivery)

- **Surface erosion**

- The rate at which water penetrates into the polymeric device is slower than the rate of erosion
 - The polymer starts eroding before water has penetrated the entire volume of the device
 - leads to erosion on the device (particle or implants) surface
 - device will become thinner or smaller over time.
 - Hydrolytically surface eroding: Polyanhydrides. Poly(orthoesters). Note this is also dependent on fabrication process and device geometry.

Factors influencing hydrolytic bio-erosion

- **Backbone hydrolysis is the most common mechanism of erosion** for currently available synthetic biopolymers.
- Main **factors determining overall rate of erosion**:
 - **Chemical stability of backbone: How labile are the linkages?**
 - Susceptibility to hydrolytic cleavage: Anhydrides > Esters > Amides → Faster erosion
 - **Hydrophobicity of the monomer: ability of water molecules to penetrate into the polymer structure.**
 - E.g. Erosion of poly(anhydrides) slows down ~ three orders of magnitude by replacing the hydrophilic monomer sebacic acid with suitable hydrophobic monomers
 - **Morphology of the polymer:** Crystalline polymers are dense and resist water penetration. Amorphous polymers with $T_g > 37^\circ\text{C}$ are at glassy state while $T_g < 37^\circ\text{C}$ are at rubber state. Glassy polymers are less permeable to water.
 - **Molecular weight, fabrication process** etc
 - **Geometry of the implanted device:** Surface to volume ratio

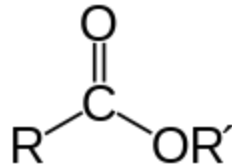
Host-induced hydrolytic processes

- **Simple hydrolysis:** Action of neutral water. E.g. polyglycolic acid erosion.
- **Ion-catalyzed hydrolysis:** Ions present in extracellular fluids (PO_4^{-3} , H^+ , OH^- , Na^+ , Cl^- , Mg^{2+} , Ca^{2+} etc.). Certain ions can catalyze hydrolytic reactions. E.g. polyester hydrolysis rate increases several folds in presence of cations
- Local **pH changes** due to inflammatory reactions have been shown to accelerate hydrolysis
- **Proteases, elastases, and other enzymes** increase the rates of hydrolysis, especially for polyesters and polyamides

Oxidative biodegradation

- Free radical initiated oxidation of polymers → significant for polyethers (PEG), polyamides etc.
 - Host induced: Activated phagocytic cells (macrophages, neutrophils etc.) produce directly or indirectly, superoxide anion, hydrogen peroxide, all of which are powerful oxidants.
 - Environment mediated: Metal ion induced oxidation (polyether urethane). Cracks on the polymer device.

Biodegradable polymers: Polyesters



- Polymers with **ester linkages between monomers**
- **Fast hydrolysis of ester bond** → **bulk degradation** with random hydrolytic scission of the polymer backbone
- Most common synthesis process: **Ring opening polymerization of the corresponding cyclic monomer** → **Catalysts (most commonly stannous octoate) are routinely used** to increase polymerization efficiency and yield high molecular weight polymers

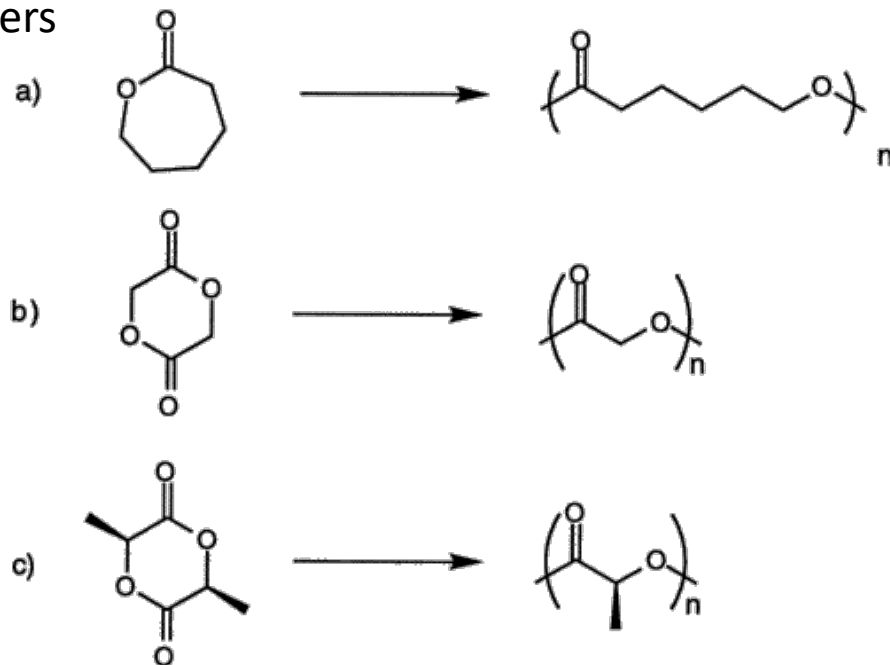
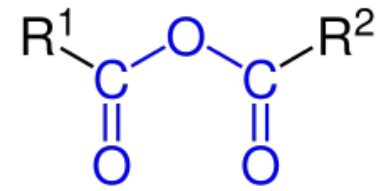


Figure: Ring-opening polymerization of selected cyclic lactones to give the following: (a) poly(-caprolactone) (PCL); (b) poly(glycolic acid) (PGA); (c) poly(L-lactic acid) (PLA).

Polyesters: PLA, PGA, PLGA

- Most extensively used biodegradable polymers
 - FDA approved as bio-absorbable sutures
 - Used extensively as particles in drug and gene delivery
 - Tissue engineering scaffolds
- Hydrolysable into lactic acid and glycolic acid
 - Lactic and glycolic acid are normal physiological products
 - Ultimately metabolized by the body into CO_2 and H_2O
 - However, **accumulation of acids creates local pH drop causing irritation**
 - Hydrolysis is catalyzed by acids and bases
 - PLA hydrolysis is slower than PGA (bulky methyl group), PLGA hydrolysis depends upon composition
 - Since Poly(L-lactic acid) is highly crystalline compared to Poly(DL-lactic acid) or PLGA it has much higher T_g and slower rate of hydrolysis.
 - Polymers with acid end groups ($-\text{COOH}$) hydrolyses faster $\rightarrow$ acid end group catalysis

Polyanhydrides



- Surface erosion can be achieved by synthesizing polymers that are hydrophobic (prevents water from entering device) yet contains highly water-sensitive linkages → polyanhydrides
- **Produces near zero-order release kinetics**
 - Constant rate of release due to surface degradation only
- **Degradation rate can be controlled by polymer composition**
 - Changing the hydrophobic monomers → aliphatic anhydrides degrade in days whereas aromatic ones can take years
- **Biocompatible**
 - Has been used in clinical trials for brain cancer patients (drug release implants)

Structures of representative polyanhydrides and their copolymers used in localized delivery applications

Polymer	Structure
<i>Aliphatic polyanhydrides</i>	
PSA	$\left[\text{C}(=\text{O})-(\text{CH}_2)_8-\text{C}(=\text{O})-\text{O} \right]_n$
<i>Aromatic polyanhydrides</i>	
P(CPV)	$\left[\text{C}(=\text{O})-\text{C}_6\text{H}_4-\text{O}-(\text{CH}_2)_4-\text{C}(=\text{O})-\text{O} \right]_n$
P(TA-IPA)	$\left[\text{O}-\text{C}(=\text{O})-\text{C}_6\text{H}_4-\text{CH}_2-\text{O}-\text{C}(=\text{O})-\text{C}_6\text{H}_4-\text{C}(=\text{O}) \right]_n$
P(CPP-IPA)	$\left[\text{C}(=\text{O})-\text{C}_6\text{H}_4-\text{O}-\text{CH}_2\text{CH}_2\text{CH}_2-\text{O}-\text{C}(=\text{O})-\text{C}_6\text{H}_4-\text{C}(=\text{O})-\text{O}-\text{C}(=\text{O})-\text{C}_6\text{H}_4-\text{C}(=\text{O}) \right]_n$
<i>Aliphatic-Aromatic Polyanhydrides</i>	
P(CPP-SA)	$\left[\text{C}(=\text{O})-\text{C}_6\text{H}_4-(\text{CH}_2)_3-\text{C}_6\text{H}_4-\text{C}(=\text{O})-\text{O} \right]_n \left[\text{C}(=\text{O})-(\text{CH}_2)_8-\text{C}(=\text{O})-\text{O} \right]_m$
P(CPM-SA)	$\left[\text{C}(=\text{O})-\text{C}_6\text{H}_4-(\text{CH}_2)_1-\text{C}_6\text{H}_4-\text{C}(=\text{O})-\text{O} \right]_n \left[\text{C}(=\text{O})-(\text{CH}_2)_8-\text{C}(=\text{O})-\text{O} \right]_m$
P(CPH-SA)	$\left[\text{C}(=\text{O})-\text{C}_6\text{H}_4-(\text{CH}_2)_6-\text{C}_6\text{H}_4-\text{C}(=\text{O})-\text{O} \right]_n \left[\text{C}(=\text{O})-(\text{CH}_2)_8-\text{C}(=\text{O})-\text{O} \right]_m$

Different polyanhydrides studied for various applications as localized drug carriers

Disease	Polymer	Delivery system	Drugs
Cancer	Ricinoleic acid based polyanhydride	Matrix	Methotrexate
	P(RA-SA)	Matrix	Cisplatin
	P(CPP-SA)	Implant	Bromodeoxyuridine
			N-(phosphonacetyl)-aspartic acid
	P(CPP-SA)	Implant	5-fluorouracil
	P(FAD-SA)	Implant	Taxol
	P(CPP-SA)	Implant	Camptothecin
	P(RA-SA)	Injectable Paste	Paclitaxel
Head and neck cancer	P(FAD-SA)	Matrix	Cisplatin, 5-FU, methotrexate, paclitaxel
Osteomyelitis	P(OA/LAD-SA)	Matrix	Gentamicin
Local anesthesia	P(FAD-SA)	Matrix	Bupivacaine HCl
	P(CPP-SA)	Implant	Dibucaine, bupivacaine
Local infection	P(DDDA-TA), P(BA-PA)	Matrix	Ciprofloxacin hydrochloride
	P(FAD-SA)	Implant	Cefazolin sodium
Restenosis	P(EAD-SA)	Implants	Heparin
Gene delivery	Photocrosslinked polyanhydride	Matrix	DNA
Glaucoma	P(CPP-SA)	Implant	Etoposide
Inflammatory bowel disease	Poly(anhydride-esters)	Microspheres	Aminosalicylates
Inflammation	PLA-PSA-PLA	Microspheres	Triamcinolone
Hormone therapy	P(FAD-SA)	Microsphere	GnRH α
Alzheimer's disease	SA copolymer	Microspheres	Bethanechol

Commercial Therapeutics

Proteins, DNA and other biomolecules are highly evolved and specific, so they make excellent drugs

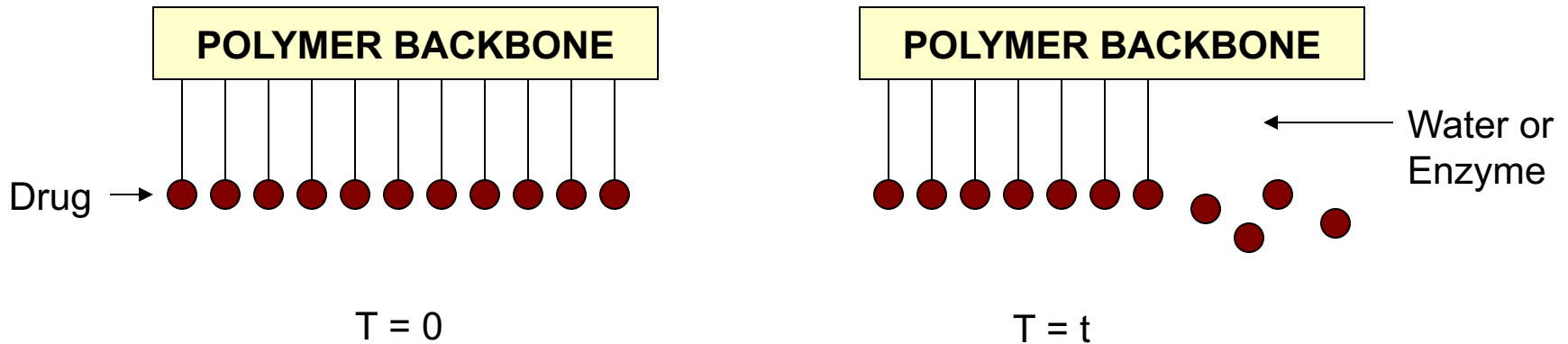
- **Insulin**
 - Helps to regulate blood glucose levels for people with diabetes.
- **Interferon- α (Intron A, Roferon)**
 - Used for the treatment of chronic hepatitis C in adults.

Biomolecules such as proteins rapidly degrade in the body by natural mechanisms

This means that in order to have a sustained affect – the patient must endure many injections

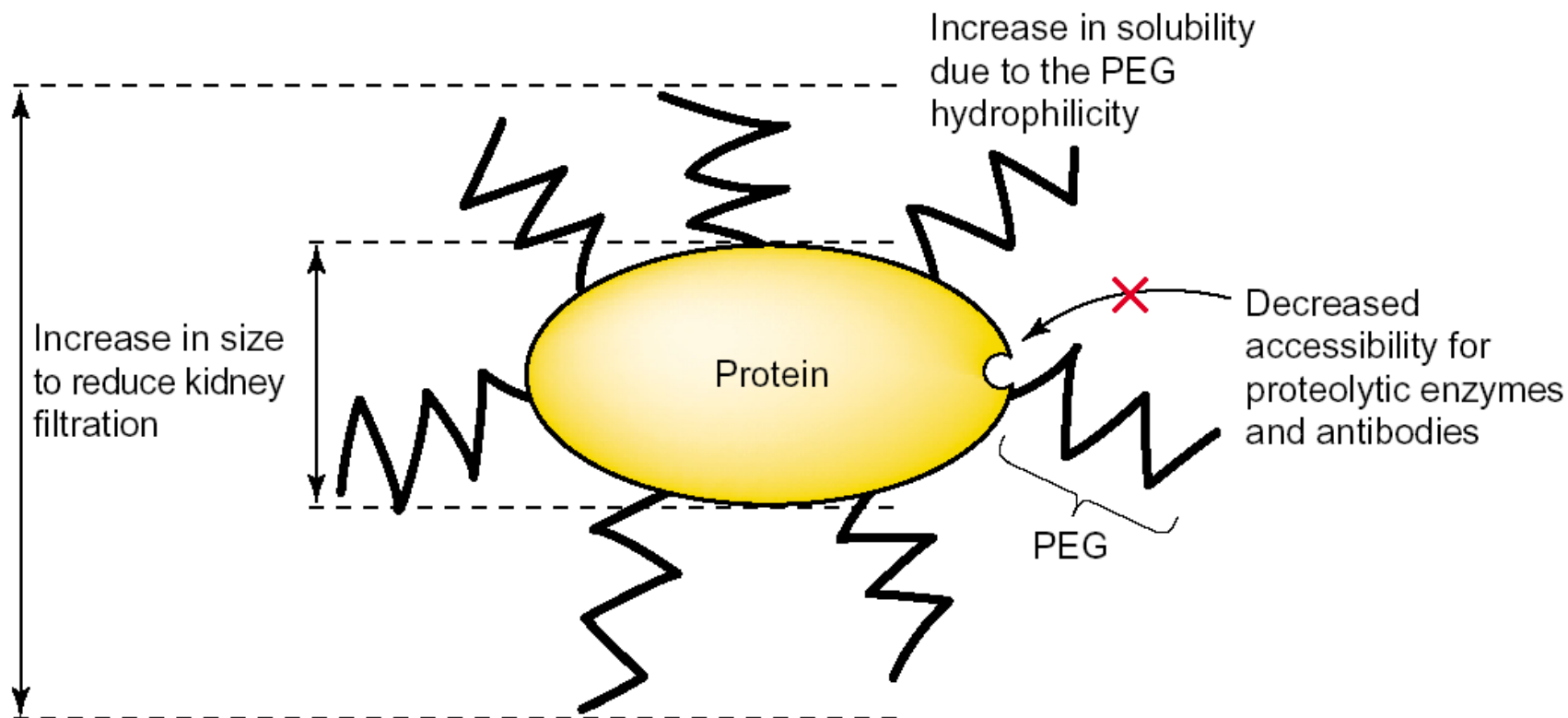


Polymer-drug conjugates: Small molecule



- Primary advantage
 - Very high drug loading
- Primary disadvantage
 - New Chemical entity

Protein-Polymer Conjugates for large molecules



Drug Discovery Today

Protein-polymer therapeutics are **nano-sized** drugs with many advantages

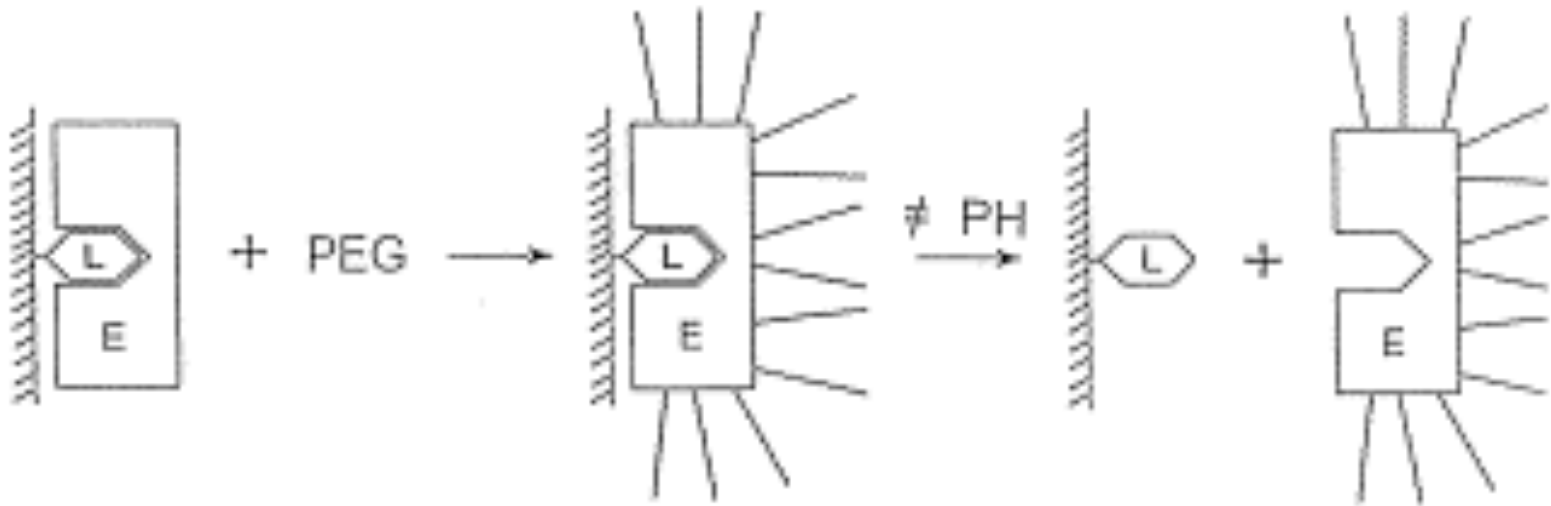
Advantages of polymer conjugates

- Modify chemical properties – Enhance Solubility
- Protection of labile drugs from biochemical degradation - Stability
- Increased half-life (body residence time) - Stability
- Reduction of immunogenicity and toxicity - Safety
- Improved targeting (to specific organs and specific cells) – Specificity and bioavailability, Controlled delivery
- Drug conjugation via degradable bonds – Controlled release
- Drug conjugation to stimuli responsive polymers – Delivery under specific conditions – Controlled release

Limitations of polymer-drug conjugates

- **Bio-molecule (drug) can loose activity during conjugation process** – careful choice of chemistry and reaction sites.
- **Purification** of un-reacted polymer, bio-molecules and any reaction products is necessary before in-vivo application
- **Sterilization** process now involves polymer stability
- Can be viewed by regulatory authorities as a **“new” drug entity** (novel chemical composition) even though the drug molecule itself is well used: could take years and years along with high cost

Active site protection

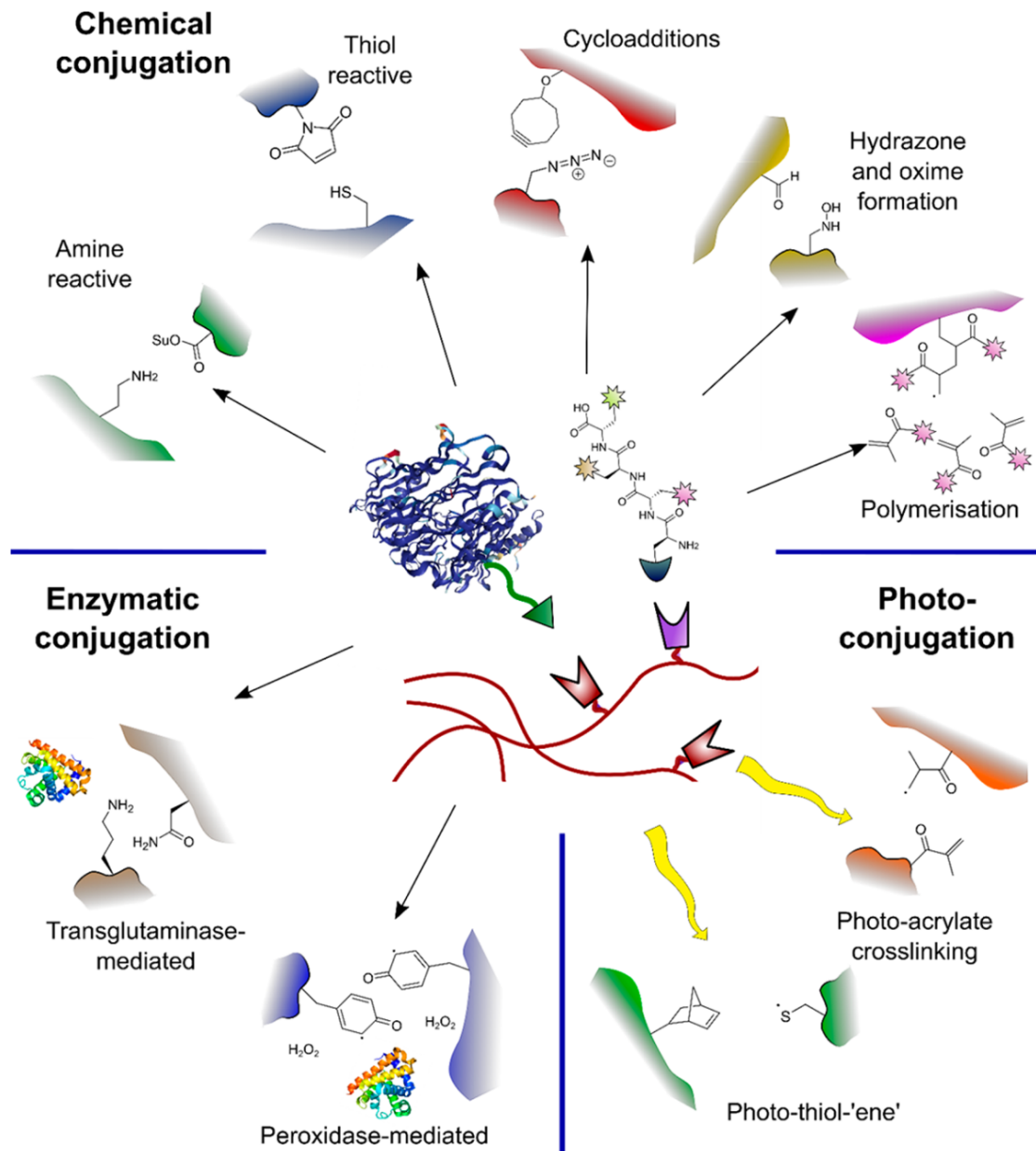


- Fig. A ligand, such as a substrate or inhibitor (L), linked to an insoluble matrix gives reversible binding with the enzyme or protein and prevents the PEGylation around the binding site. The PEGylated macromolecule is released from the ligand by any condition that reverses the binding such as pH, salts, solvents.

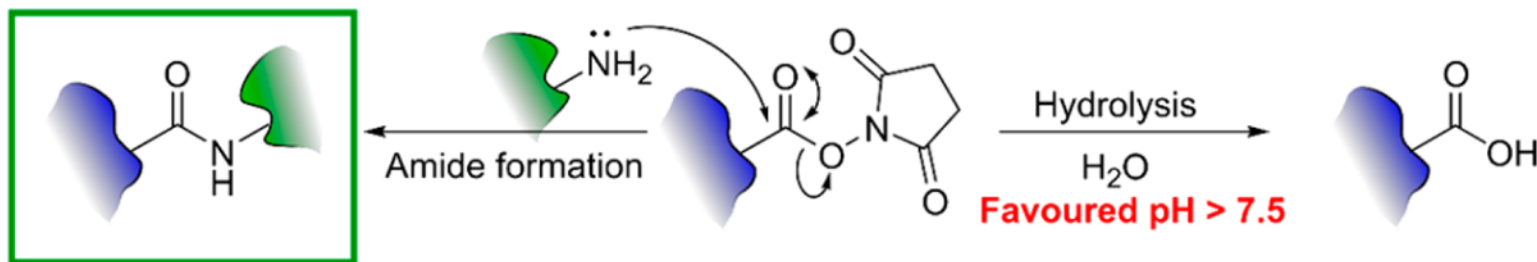
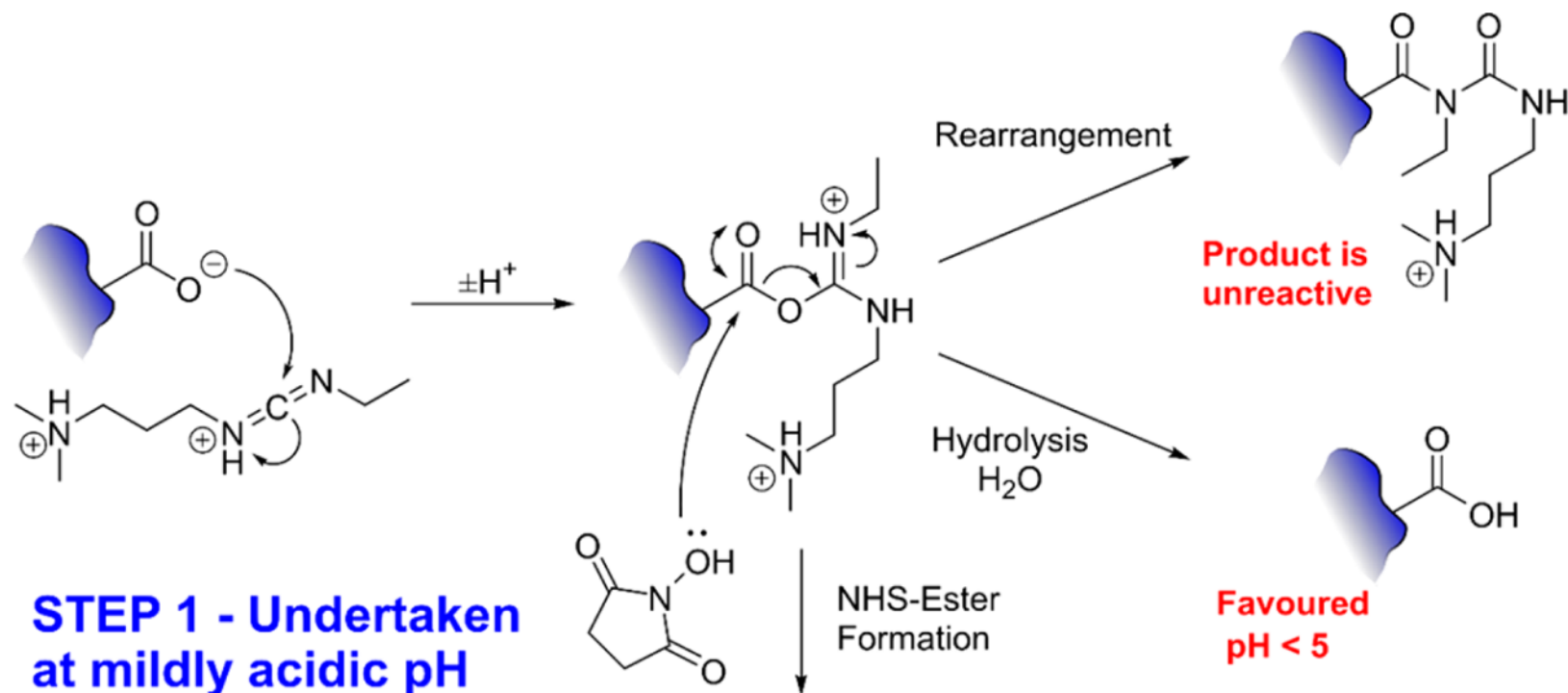
Polymers in drug conjugation

- Mostly synthetic polymers are used for conjugation owing to their controllable structure and properties
 - Exceptions, natural polysaccharides e.g. Dextrans
- Usually water soluble (hydrophilic), highly biocompatible polymers containing reactive functional groups are used
- Most commonly used polymers
 - Polyethylene glycol (PEG) and derivatives
 - N-(2-hydroxypropyl) methyl acrylamide (HPMA)
 - Dextrans and other polysaccharides
 - Poly(amino acids)
 - Stimuli-sensitive polymers etc.

Biomolecule-Biomaterial Conjugation



Biomolecule-Biomaterial Conjugation: EDC coupling



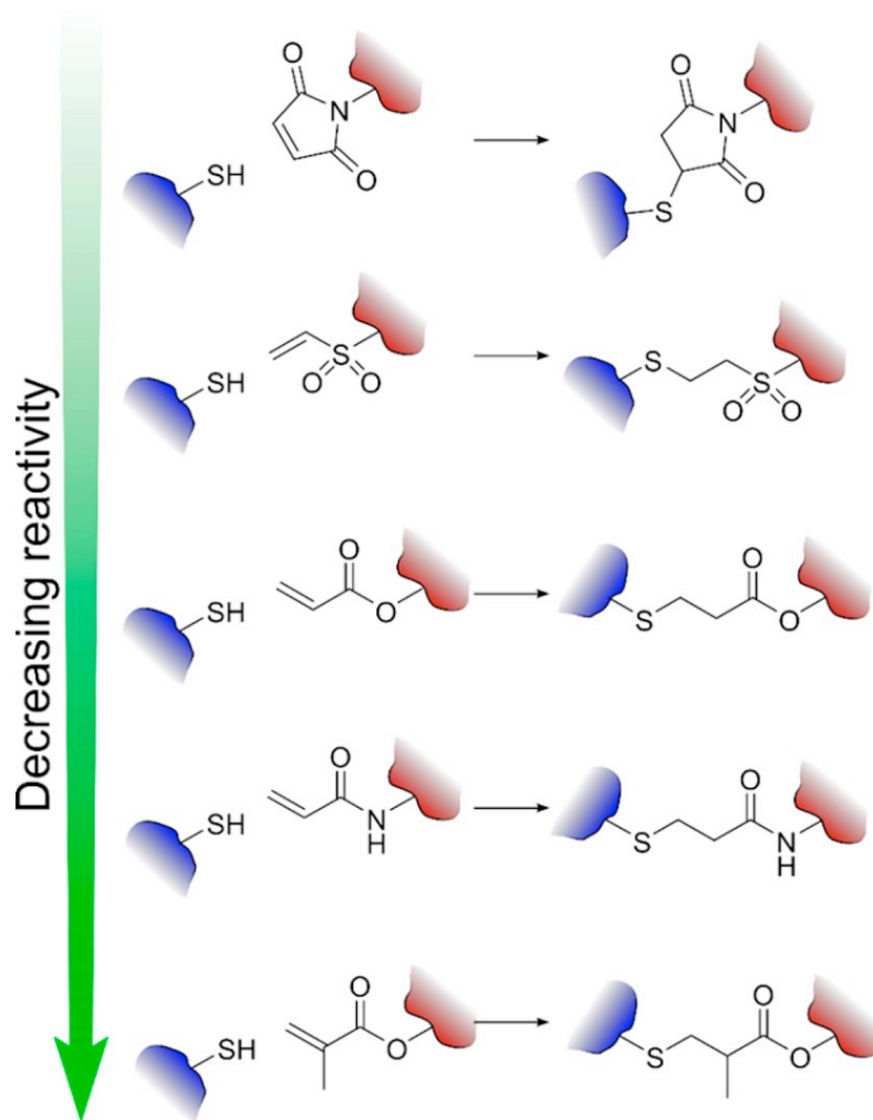
STEP 2 - Undertaken at neutral pH

Biomolecule-Biomaterial Conjugation: Thiols – Michael Addition

Less than 2% proteins have surface thiols and hence reactions are highly site specific

But at the same time they are usually involved in active site or control protein function so can be counterproductive

Within the most widely utilized acceptors in the biomaterial field, a trend of reactivity can be generalized as maleimides > vinyl sulfones > acrylates > acrylamides > methacrylates.



Functional groups on the drug:

Small molecules

- Small molecule drugs
 - Normally conjugation is performed with nucleophilic residues on the drug molecule
 - Hydroxyl (OH), Amine (NH₂) or Carboxyl (COOH) groups
 - The reactive group must not be of critical relevance for drug activity

Functional groups on the drug:

Proteins and Peptides

- Amines (Lysine, Arginine, 5' end amine, Histidine): Most widely used residue for conjugation. 10% of overall amino acids is lysine and only few are involved in the active site.
- Sulfhydryls (Cysteine): Highly reactive, but present only in few cases, and also responsible for dimer, trimerizations.
- Carboxyls (Aspartic and Glutamic acids):
- Sugar moieties in glycoproteins or polysaccharides (carrying OH, amine or aldehydes) are also used for conjugations

Reactive groups in polymers

- Polymers used in drug conjugation:
 - PEG, HPMA, Dextrans, Poly(L-lysine), poly(amino acids), polyvinyl pyrrolidone (PVP), etc.
- Primary reactive groups present in polymers:
 - OH, NH₂, COOH that are already present or derivatized
- Three distinct strategies for conjugation:
 - Reaction of the drug with functional groups present on the polymer chain
 - Reaction of polymer with an intermediate followed by drug attachment
 - Reaction of drug with an intermediate followed by polymer attachment

Example: Pre-derivatization of Dextrans

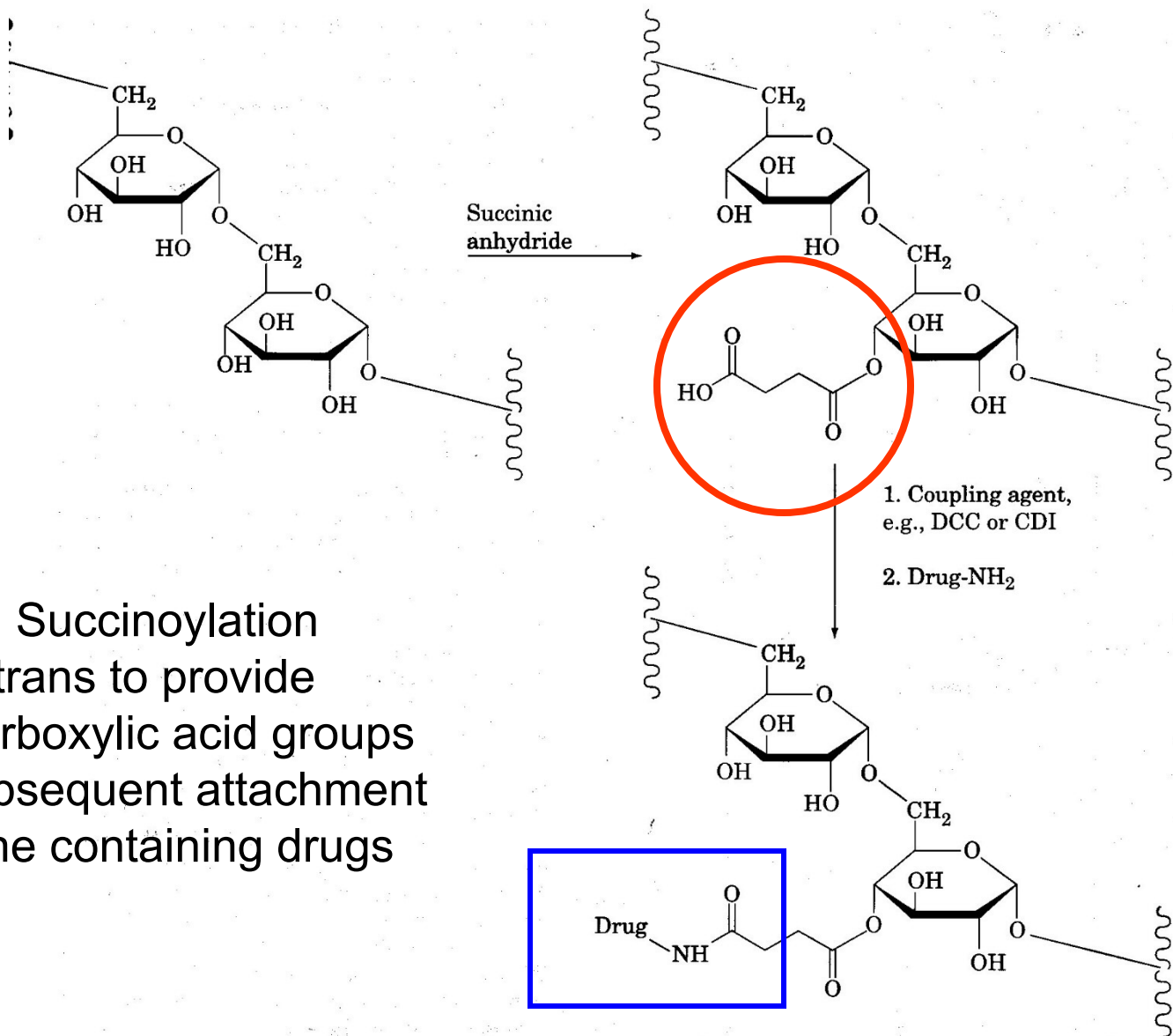
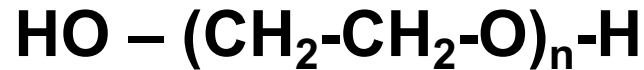


Figure: Succinylation of Dextrans to provide free carboxylic acid groups For subsequent attachment of amine containing drugs

Poly(ethylene glycol) [PEG]



- Very hydrophilic, highly biocompatible polymer
 - The backbone of the polymer is also highly flexible
 - Both these properties contribute to **“molecular windshield wiper”**
- Soluble in both aqueous and organic solvents
 - Well developed conjugation chemistries
- Non-toxic, non-immunogenic, can be produced under GMP (good manufacturing practice) conditions and FDA approved

Branched PEGylation

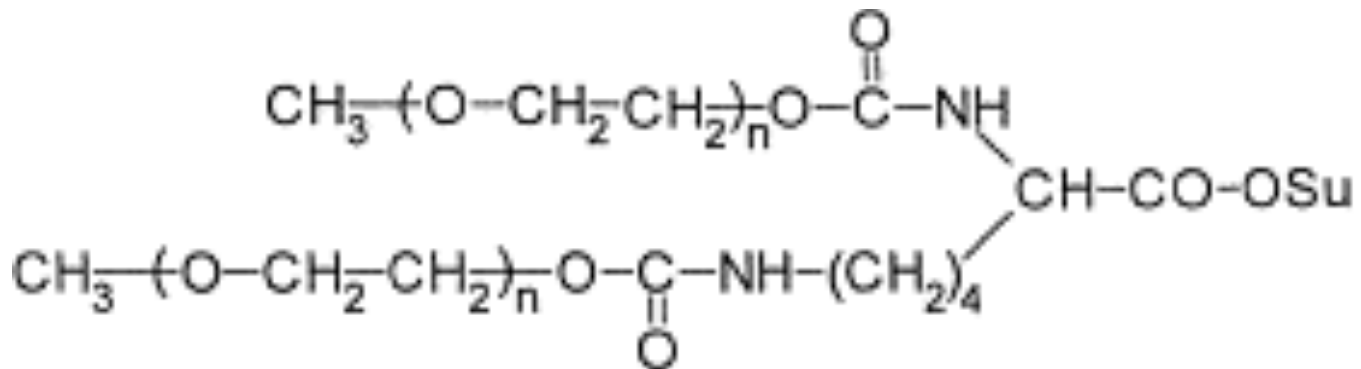


Fig. **Branched PEG (PEG2)**: Lysine is the amino acid spacer between two PEG molecules and the activated carboxylate. This PEG offers the advantages of higher protection of protein surface for single point of conjugation, as compared to the linear form. Generally, it conveys to its products higher retention in blood, lower immunogenicity and decreased enzyme inactivation during the conjugation.

Branched PEGylation

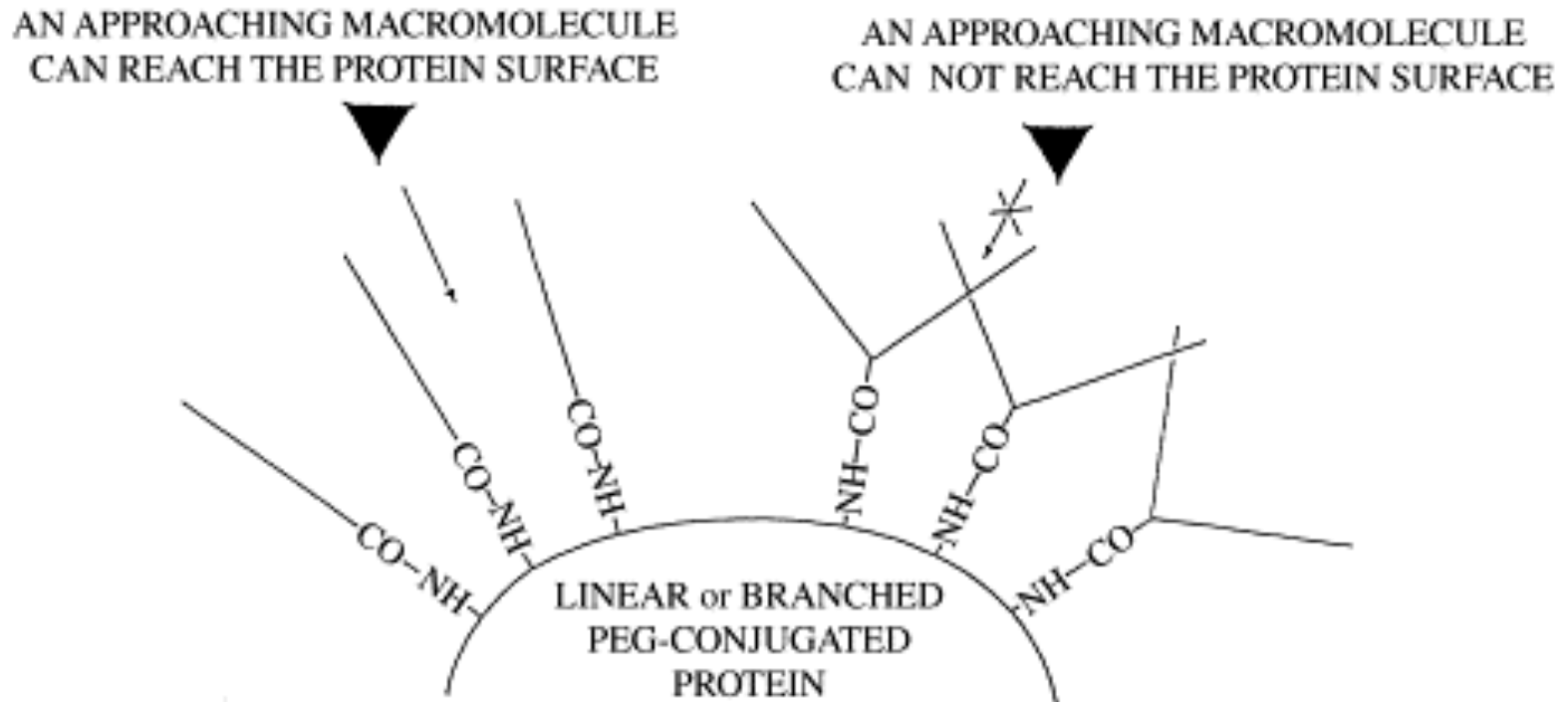
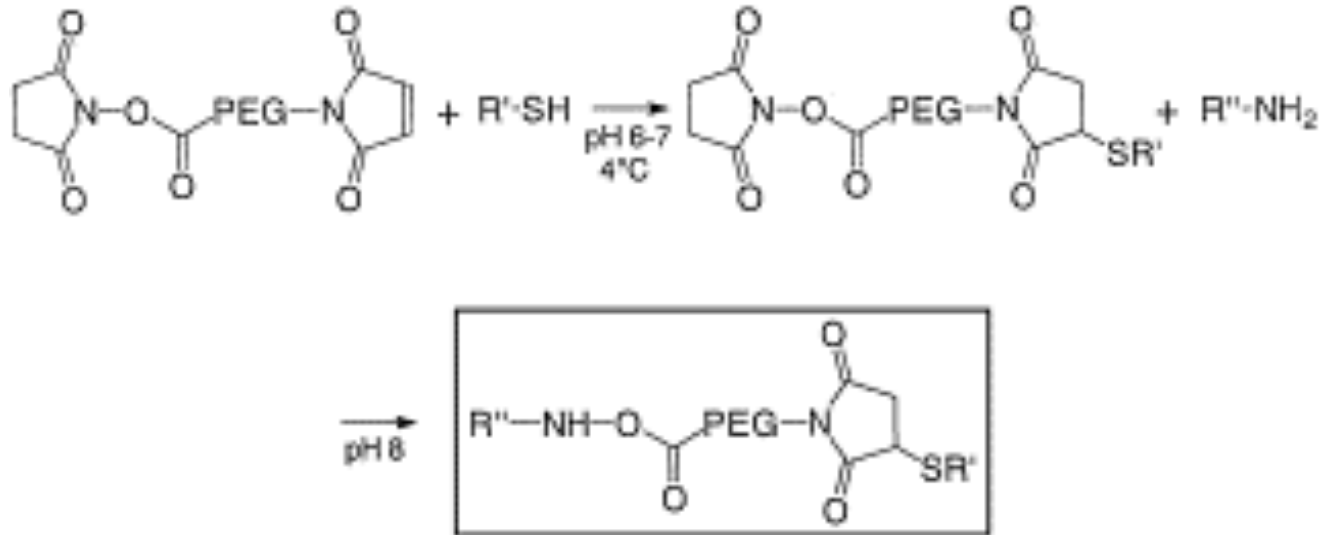


Fig. The "umbrella-like" structure of branched PEG explains the higher capacity in rejecting approaching molecules or cells as compared to linear PEG of the same size.

Linking multiple molecules



- Fig. -OSu- -maleimide-PEG: This is a typical cross-linking reagent that may reacts in a sequential order with different groups in protein, taking advantage of different pH and temperatures. With this strategy two different macromolecules may be stoichiometrically linked together.

PEGylation: solubility and clearance

- Water solubility of Taxol (a widely used chemotherapy drug) with or without PEG attachment

Mw of PEG	Solubility (mg/ml)
No PEG	~ 0
5,000	660
20,000	200
40,000	125

- Renal clearance and enzymatic degradation of proteins can be decreased thereby **increasing serum residence time** (SOD: Super-oxide dismutase)

Protein ^c	Half-life
SOD	0.08 ^a
SOD-PEG 1900 Da	1.5 ^a
SOD-PEG 5000 Da	11.0 ^a
SOD-PEG 72000 Da	36.0 ^b

Pharmacokinetics of PEGylated drugs



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Pharmacokinetic and biodistribution properties of poly(ethylene glycol)–protein conjugates

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Table 1

Physico–chemical and pharmacokinetic properties of PEGs: correlation among molecular weight, molecular size, protein molecular weight equivalence, permanence in the circulation after intravenous administration

PEG molecular weight (kDa)	PEG molecular size (nm) ^a	Protein molecular weight equivalence (gel permeation) ^b	Half time in blood (min) ^a
PEG 5 kDa	2	Cytochrome <i>c</i> (13 kDa)	18
PEG 10 kDa	–	Superoxide dismutase (32 kDa)	–
PEG 12 kDa	5	–	–
PEG 20 kDa	7	–	180
PEG 32 kDa	8	IgG (150 kDa)	–
PEG 40 kDa	10	–	–
PEG 48 kDa	11	–	840
PEG 73 kDa	–	Thyroglobulin (620 kDa)	–
PEG 180 kDa	24	–	1200
PEG 220 kDa	27	–	–

^a The data were obtained from the paper of Nakaoka et al. [16].

^b The data were obtained from the paper of Sherman et al. [13].

Properties of PEGylated drugs: Low immunogenicity

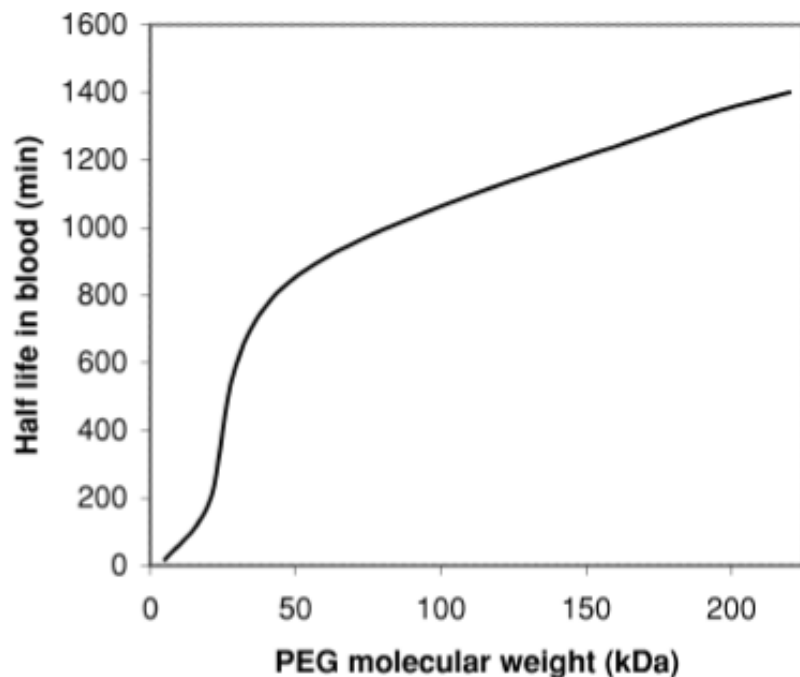


Fig. 1. Correlation plot between PEG molecular weight and half life in blood after intravenous administration. The data were obtained by the papers of Yamoka et al. [14] and Nakaoka et al. [16].

Table 2

Role of PEG shape on uricase immunogenicity^a

	Immunogenicity (%) ^b	
	IgG	IgM
Native uricase	100	100
Linear 5000 PEG uricase	0.2	0.3
Branched 10000 PEG uricase	0.2	0

^a The results were obtained from the paper of Caliceti et al. [45].

^b The immunogenicity study was carried out by mouse immunization with equimolar amount of PEG-enzyme or native enzyme. Uricase was modified with PEG to yield the same degree of derivatized amino groups. The protein immunogenicity of the various uricase forms was calculated as maximal serum dilution to reach the same enzyme-linked immunosorbent assay (ELISA) response and expressed as percentage with respect of the value obtained with the native enzyme.

PEGylation: solubility and clearance

- Water solubility of Taxol (a widely used chemotherapy drug) with or without PEG attachment

Mw of PEG	Solubility (mg/ml)
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5,000	660
20,000	200
40,000	125

- Renal clearance and enzymatic degradation of proteins can be decreased thereby **increasing serum residence time** (SOD: Super-oxide dismutase)

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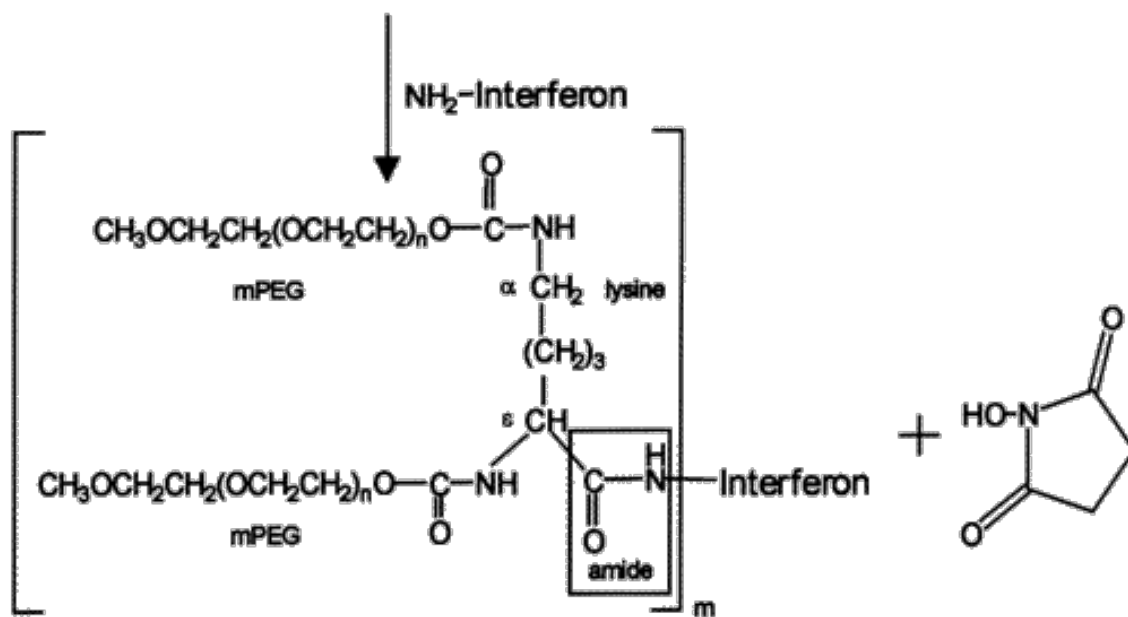
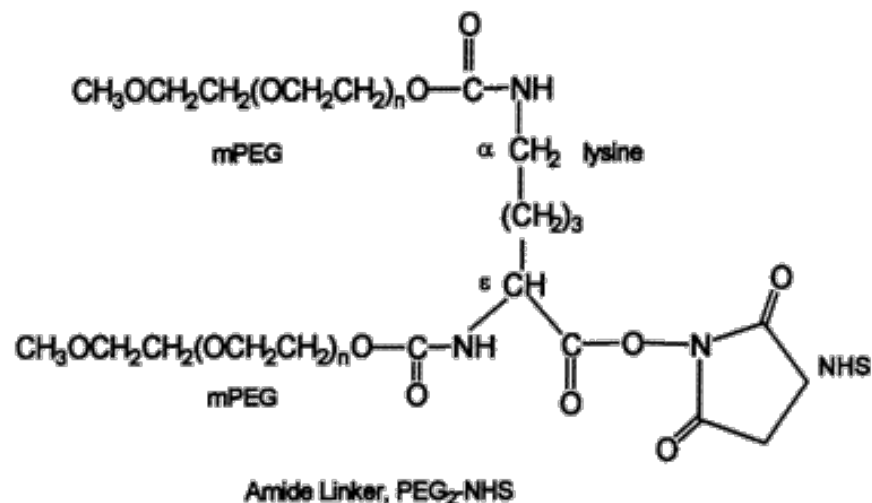
PEGylation: Specific Example

Interferon alpha -2a (IFN α -2a), Hoffmann-La Roche

IFN alpha is a cytokine with potent anti-viral and anti-tumor activities

- **Interferon alpha:**
 - **Terminal half-life** of interferon: 4 to 8 h, following intravenous (i.v.) or subcutaneous (s.c.) administration.
 - Little or no interferon is detected at 24 h
 - Necessitating frequent dosing of interferon. Certain oncological indications require daily dosing, while viral indications may require three times weekly dosing
 - Interferon treatment for these diseases: periods of several months to a year or longer

PEGylation of IFN alpha: Chemistry



where $m=1$ and $n=420-510$

PEG2-IFN: Identification after conjugation

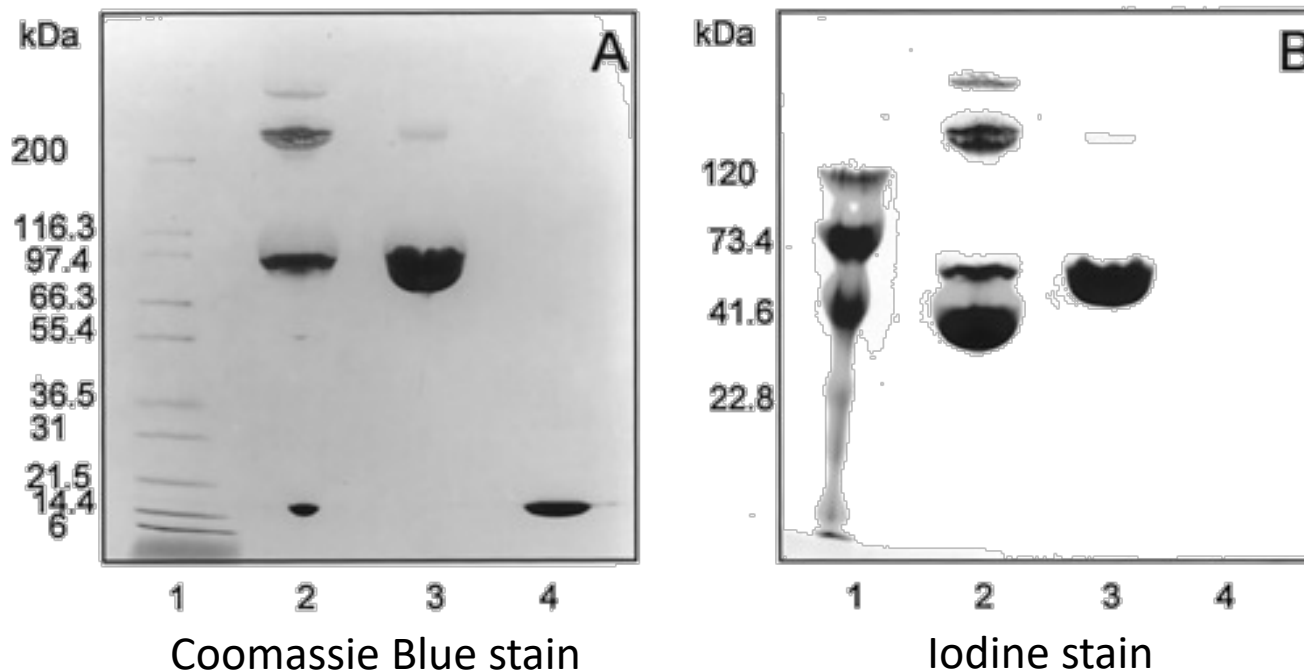


Figure: SDS-PAGE analysis. (A) Specifically stained for protein with Coomassie blue. Lanes: 1, molecular weight marker proteins; 2, pegylation reaction mixture; 3, purified PEG2-IFN; and 4, interferon -2a. (B) Specifically stained for PEG with iodine. Lanes: 1, molecular weight marker PEGs stained by iodine; 2-4, same as in Figure 2A. Note that lane 4 containing interferon -2a in gel B is not

PEGylation: Specific Example

Interferon alpha -2a (IFN α -2a), Hoffmann-La Roche

IFN alpha is a cytokine with potent anti-viral and anti-tumor activities

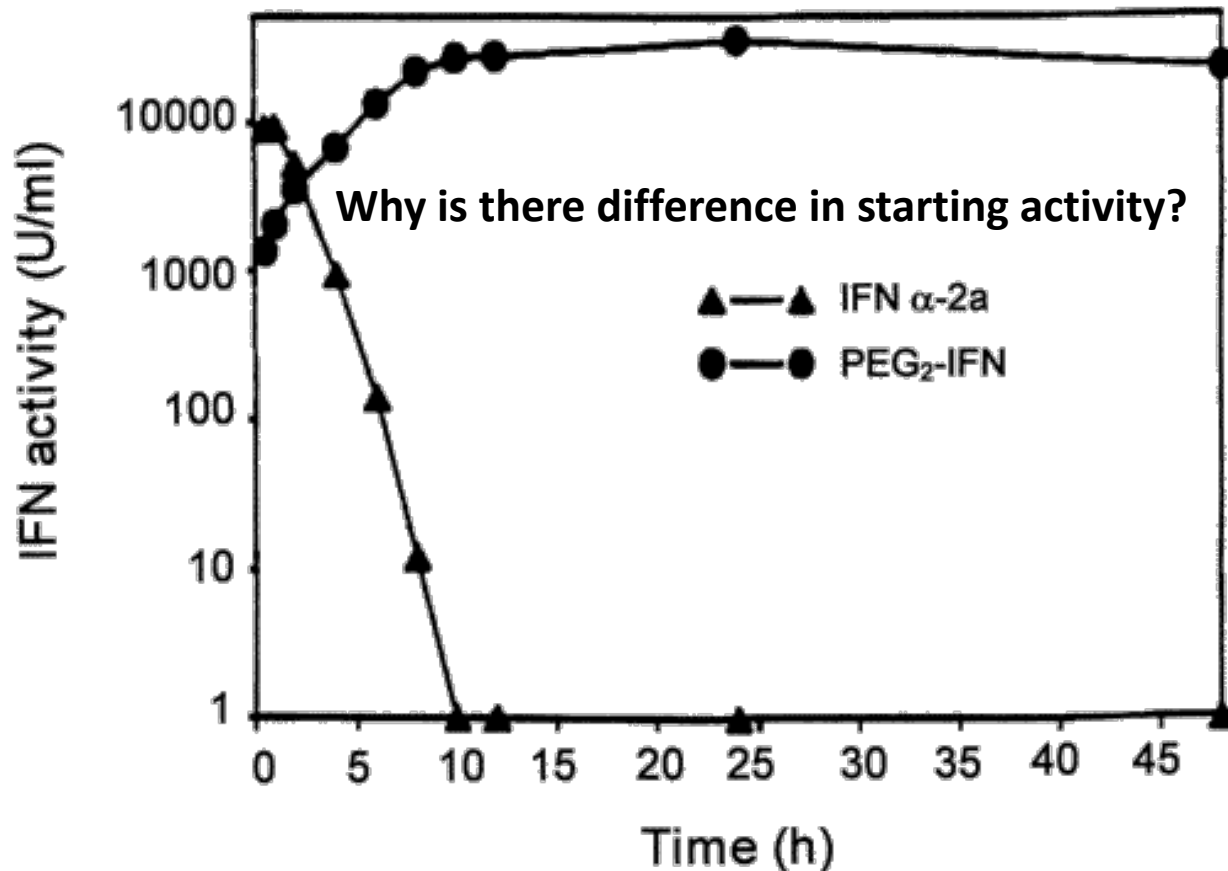


Figure Mean serum activity versus time of interferon -2a and PEG₂-IFN after subcutaneous injection in rats. PEG₂-IFN has a 51-h half-life compared to 0.7-h for interferon -2a concomitant with mean plasma residence time of 80 and 1.6 h, respectively.

Protein-Polymer Conjugates on Market

Name (company)	Polymer carrier	Drug	Indication(s)	Year approved*	Refs
Adagen (Enzon Pharmaceuticals)	PEG	ADA	ADA-SCID	1990	12
SMANCS (Astellas Pharma)	Poly(styrene-co-maleic acid)	Neocarzinostatin	Liver and renal cancer	1993 (Japan)	260
Oncaspar (Enzon Pharmaceuticals)	PEG	L-asparaginase	Acute lymphoblastic leukaemia	1994	35
PegIntron (Merck & Co.)	PEG	Interferon $\alpha 2b$	Hepatitis C	2001	261
Pegasys (Genentech)	PEG	Interferon $\alpha 2a$	Hepatitis B and hepatitis C	2002	262
Neulasta (Amgen)	PEG	G-CSF	Chemotherapy-induced neutropenia	2002	263
Somavert (Pfizer)	PEG	HGH receptor antagonist	Acromegaly	2003	264
Macugen (Bausch & Lomb)	PEG	Anti-VEGF aptamer	Neovascular age-related macular degeneration	2004	265
Mircera (Roche)	PEG	Epoetin beta	Anaemia associated with chronic kidney disease	2007	38
Cimzia (UCB Pharma)	PEG	Anti-TNF Fab'	Crohn's disease, rheumatoid arthritis, psoriatic arthritis and ankylosing spondylitis	2008	266
Krystexxa (Horizon Pharma)	PEG	Uricase	Chronic gout	2010	267
Plegridy (Biogen)	PEG	Interferon $\beta 1a$	Relapsing multiple sclerosis	2014	268
Movantik (AstraZeneca)	PEG	Naloxone	Opioid-induced constipation	2014	94
Adynovate (Baxalta)	PEG	Factor VIII	Haemophilia A	2015	269
Palynziq (BioMarin)	PEG	Phenylalanine ammonia lyase	Phenylketonuria	2018	270
Jivi (Bayer)	PEG	Factor VIII	Haemophilia A	2018	271

ADA, adenosine deaminase; Fab', antigen-binding fragment; G-CSF, granulocyte colony-stimulating factor; HGH, human growth hormone; PEG, polyethylene glycol; SCID, severe combined immunodeficiency; TNF, tumour necrosis factor; VEGF, vascular endothelial growth factor. *Approved by the US Food and Drug Administration unless otherwise stated; biosimilar versions of approved drugs are omitted.

Sterilization and storage

- Need to **minimize premature polymer degradation:**
Storage stability
 - Hydrolysis by absorption of moisture during storage
 - Air sealed, moisture resistant packaging
 - Low temperature storage
 - Hydrolysis during fabrication and processing
- **Sterilization before use in biomedical applications**
 - Most common methods are gamma radiation and exposure to ethylene oxide
 - Choosing the lesser of the two evils
 - Gamma radiation can significantly degrade polymer backbone, especially polyesters
 - Ethylene oxide is highly toxic
 - Considerable industrial effort has been put into this problem
 - Clean room processing and solution filtrations

Controlled drug release

- **Diffusion-controlled systems:**
 - Reservoir devices
 - Monolithic (matrix) devices (non-erodible)
- **Solvent-controlled systems:**
 - Osmotically controlled devices
 - Swelling-controlled devices
- **Chemically-controlled systems:**
 - Drug covalently attached to the polymer backbone
 - Drug in a core surrounded by a bio-erodible rate-controlling membrane.
 - Drug homogeneously dispersed in a bio-erodible polymer

Diffusion Controlled Systems

First law of diffusion: Fick's Law

Diffusion is defined as the movement of solute molecules from a higher to a lower concentration gradient.

“random walk”

molecules are continually colliding with each other

The diffusive flux, i.e. the mass per unit time of solute movement is mathematically expressed as :

$$J = -D \frac{\partial C}{\partial x} \dots\dots\dots(1)$$

Where $\delta C/\delta x$ is the concentration gradient in the direction of solute movement

And D is the constant of proportionality and is defined as the **Diffusion constant or Diffusion coefficient**

From a “random walk” derivation D is related to the **root mean square displacement** and time interval (t) by $x_{rms} = \sqrt{2Dt}$

If a particle takes time T to diffuse L mm. how long will it take to diffuse 2L mm?

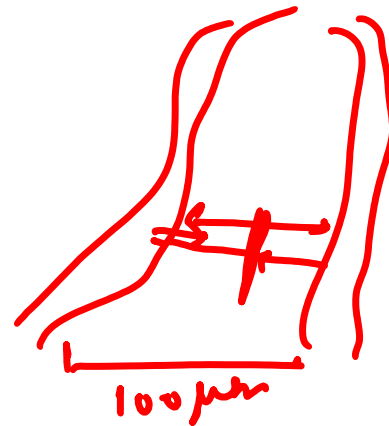
- Find how much time it will take for a protein with diffusion coefficient $5 \times 10^{-7} \text{ cm}^2/\text{sec}$ to move the root mean square distance of 1.5m.

- Find how much time it will take for a protein with diffusion coefficient $5 \times 10^{-7} \text{ cm}^2/\text{sec}$ to move the root mean square distance of 1.5m.

$$\begin{aligned}
 x_{\text{rms}} &= \sqrt{2Dt} \Rightarrow x_{\text{rms}}^2 = 2Dt \Rightarrow t = \frac{x_{\text{rms}}^2}{2D} \\
 t &= \frac{1.5 \times 1.5 \text{ m}^2}{2 \times 5 \times 10^{-7} \frac{\text{cm}^2}{\text{sec}}} = \frac{1.5 \times 1.5 \text{ m}^2 \cdot \text{sec}}{10^{-6} \text{ cm}^2} = \frac{1.5 \times 1.5 \times 10^{10} \text{ years}}{24 \times 3600 \times 365} > 100 \text{ years} \\
 t &= 1.5 \times 1.5 \times 10^{10} \text{ sec} =
 \end{aligned}$$

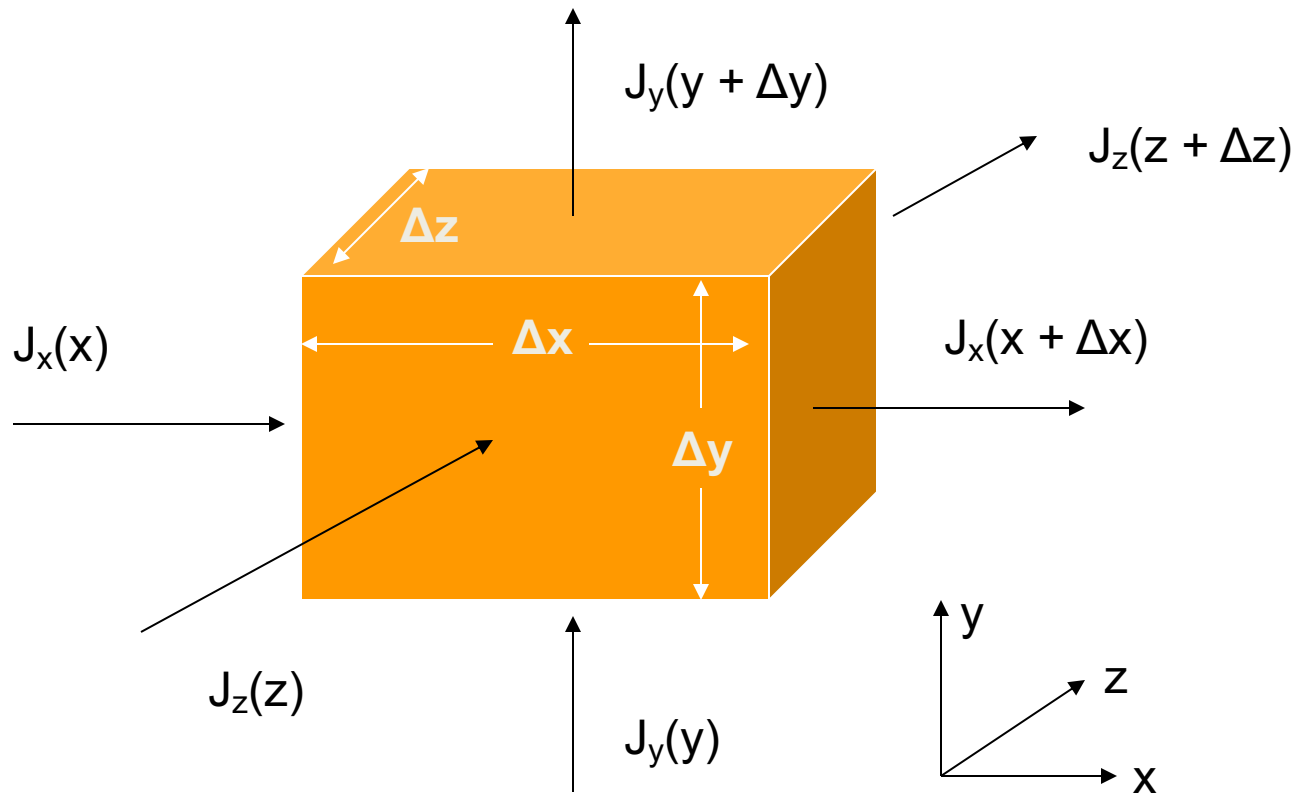
Circulatory System →

$$t = 1 \times 10^{-3} \times 10^{10} \text{ sec} = 100 \text{ sec}$$



$$\begin{aligned}
 0.1 \text{ mm} &= 0.1 \times 10^{-3} \text{ m} \\
 &= 1 \times 10^{-4} \text{ m} \\
 &= 10^{-8}
 \end{aligned}$$

Mass conservation: Second law of diffusion



General Balance Equation = (in – out) + generation = Accumulation

Mass conservation: Second law of diffusion

From the law of conservation of mass:

$$-[J_x - J_{x+\Delta x}]\Delta y\Delta z - [J_y - J_{y+\Delta y}]\Delta x\Delta z - [J_z - J_{z+\Delta z}]\Delta x\Delta y + \psi(a) = \frac{\partial C}{\partial t} \Delta x\Delta y\Delta z$$

Dividing by $\Delta x\Delta y\Delta z$ and taking the limit of the differential volume $\rightarrow 0$, we get

$$-\frac{\partial J_x}{\partial x} - \frac{\partial J_y}{\partial y} - \frac{\partial J_z}{\partial z} + \psi(a) = \frac{\partial C}{\partial t} \dots\dots\dots(2)$$

Combining equation (1) and (2) we get:

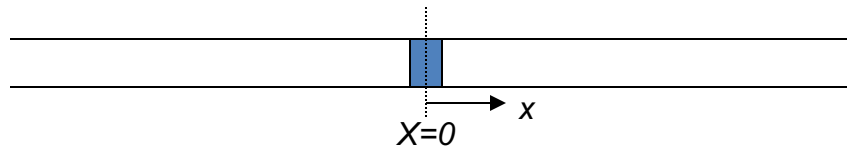
$$D\left[\frac{\partial^2 C}{\partial x^2} + \frac{\partial^2 C}{\partial y^2} + \frac{\partial^2 C}{\partial z^2}\right] + \psi(a) = \frac{\partial C}{\partial t} \dots\dots\dots(3)$$

No solute elimination or generation

Since $\Psi(a) = 0$, Equation 3 simplifies to:

$$D\nabla^2 C = \frac{\partial C}{\partial t}$$

Example 1: Rectangular coordinates, one dimension



N molecules of A injected in volume δx at $t=0$

$$D_A \frac{\partial^2 C_A}{\partial x^2} = \frac{\partial C_A}{\partial t}$$

The problem is subjected to the following initial conditions:

$$c(x, t) = N\delta(x) \quad -\infty < x < \infty, \quad t = 0$$

$$c(x, t) = 0 \quad x \rightarrow \infty, \quad t > 0$$

$$c(x, t) = 0 \quad x \rightarrow -\infty, \quad t > 0$$

Where $\delta(x)$ is the impulse function or dirac delta function centered at $x = 0$

Example1: Continued

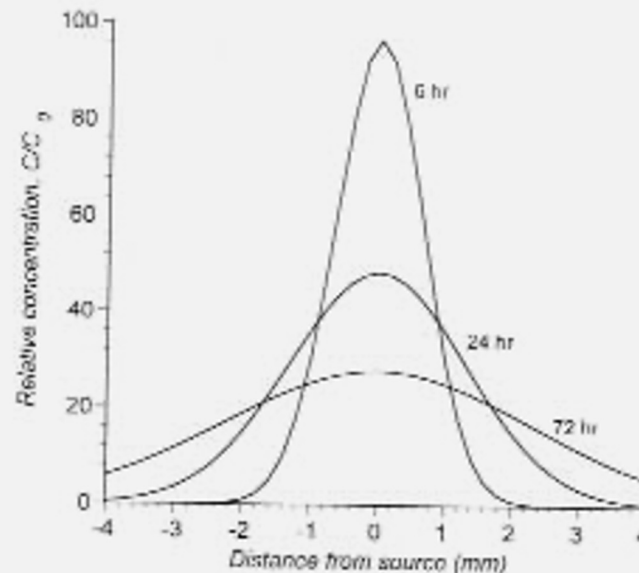
This equation can be solved using Laplace transform techniques yielding:

$$C_A(x,t) = \frac{A}{\sqrt{t}} e^{\frac{-x^2}{4D_A t}}$$

Where A is the constant of integration which can be calculated by integrating $C(x,t)$ with respect to x and equating it with the total number of diffusing molecules.

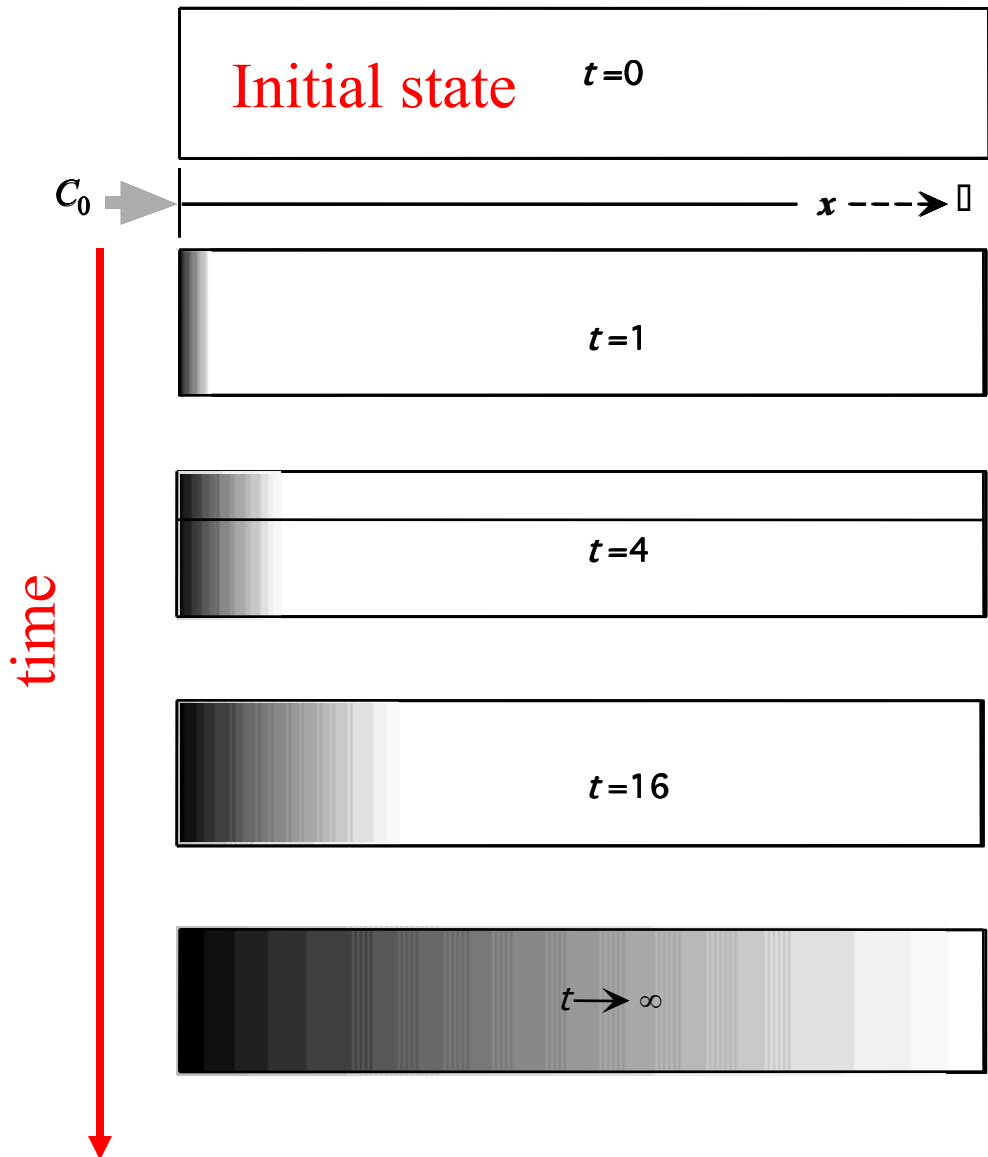
Figure 3.5

Concentration profiles for diffusion from a point source. When a concentrated bolus of solute is deposited within a small region of an infinitely long cylinder, as shown in Figure 3.4a, the molecules slowly disperse along the axis of the cylinder. The curves shown here are realization of Equation 3-34 for a solute with $D_A = 10^{-7} \text{ cm}^2/\text{s}$, $R = 0.1 \text{ cm}$, $N = N_{AV}$, and $t = 6, 24$, and 72 h .



Example 2: Diffusion from a constant source

Linear diffusion
into a semi-
infinite medium



Example 2: Diffusion from a constant source

Diffusion profile on one side of a planar boundary after suddenly raising the concentration on the other side, then keeping the source constant

This problem is subjected to different initial conditions:

$$c(x, t) = 0; \quad \text{for } 0 < x < \infty; \quad t = 0$$

$$c(x, t) = c_0; \quad \text{for } x = 0; \quad t > 0$$

The solution involves the same equation:

$$D_A \frac{\partial^2 C_A}{\partial x^2} = \frac{\partial C_A}{\partial t}$$

The analytical result is given by:

$$\frac{C_A}{c_0} = \operatorname{erfc}\left(\frac{x}{2\sqrt{D_A t}}\right)$$

Where $\operatorname{erfc}()$ is the error function complement and is given by :

$$1 - \operatorname{erf}(z) = \frac{2}{\pi} \int_0^z \exp(-\eta^2) d\eta$$

Example 2: Diffusion from a constant source

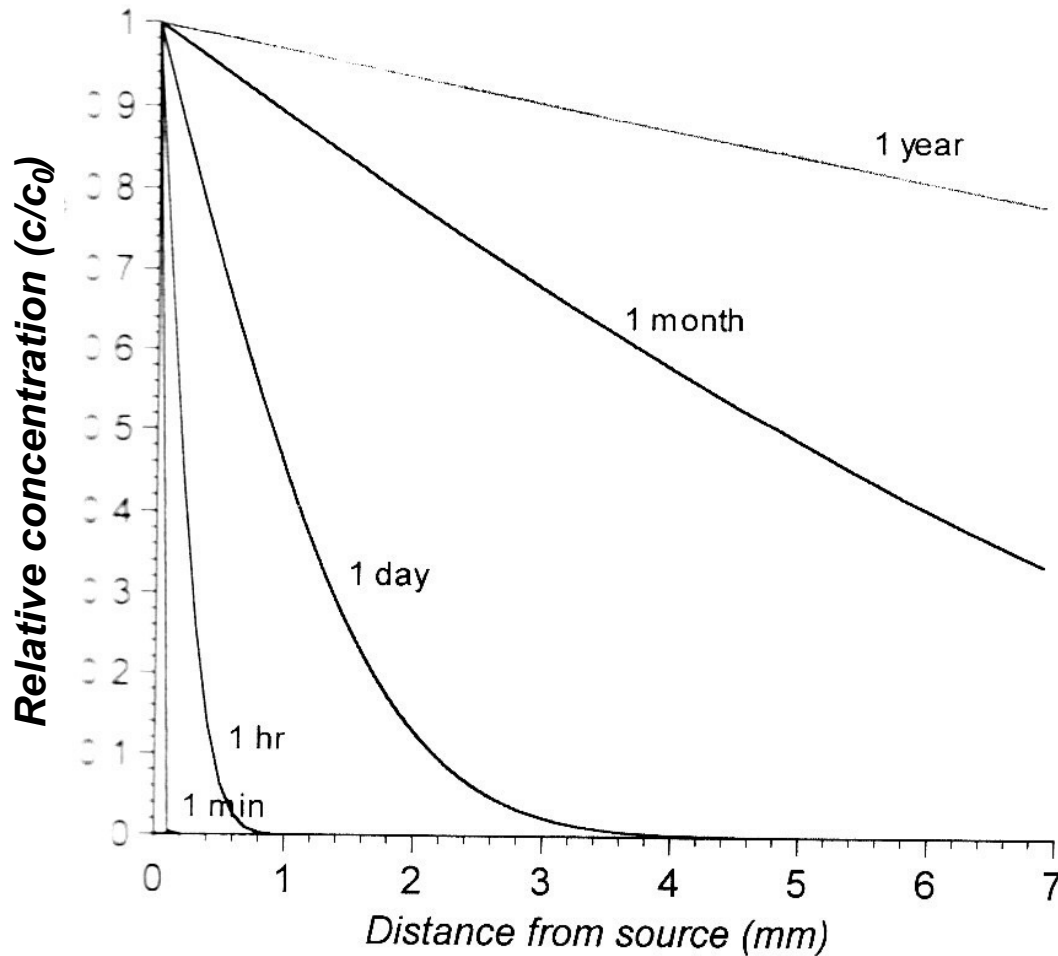


Figure 3.6

Concentration profiles for diffusion in a semi-infinite medium. When the concentration of solute is increased within one region of an unbounded space, as shown in Figure 3.4d, the molecules slowly penetrate into the adjacent region. The curves shown here are realizations of Equation 3-37 for a solute with $D_A = 10^{-7} \text{ cm}^2/\text{s}$, $R = 0.1 \text{ cm}$, $N = N_{Av}$.

Diffusion under different conditions

- The **general form of the diffusion equation remains same**
- **Diffusion constant** (coefficient D) is calculated or modified depending on physical and chemical constraints of the diffusing medium

Stokes-Einstein equation:

Diffusion of a spherical particle of radius a in a solution with viscosity μ . This equation is only valid where the **diffusing particle is large compared to surrounding solvent molecules (diffusion in aqueous medium).**

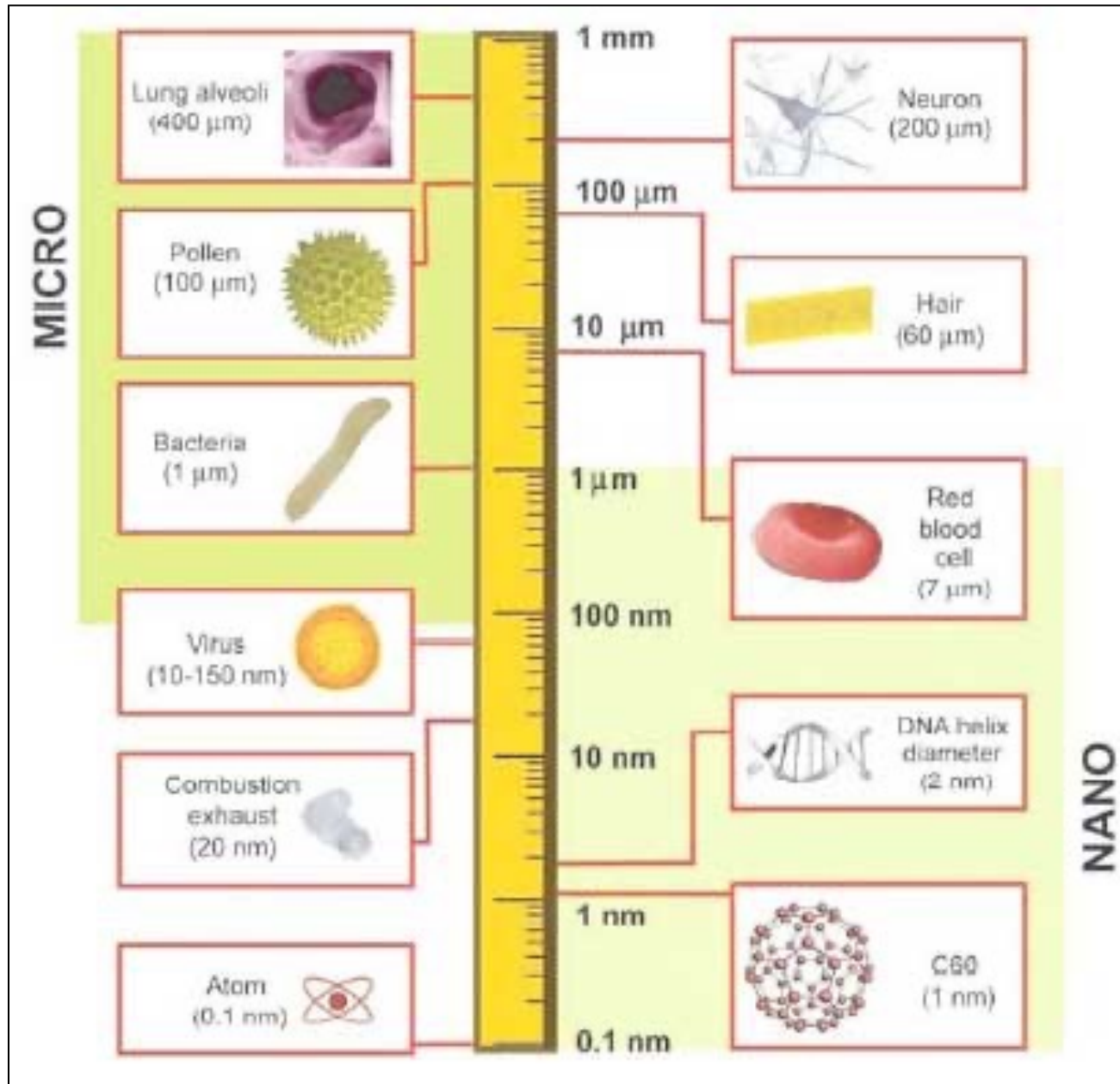
$$D_A = \frac{k_B T}{6\pi\mu a}$$

k_B is Boltzman's constant

T is the temperature in K

When "a" is measured from known D_A , it is called the hydrodynamic radius

Size scale



Protein Size

- No substantial pockets and almost no water molecules in the protein interior
- Hence, proteins are rigid structures, with a Young's modulus similar to that of Plexiglas
- All proteins have approximately the similar density, about 1.37 g/cm³

Find the relation between radius (nm) and molecular weight (Da) (assume spherical shape)

$$V(\text{nm}^3) = \frac{(0.73 \text{ cm}^3/\text{g}) \times (10^{21} \text{ nm}^3/\text{cm}^3)}{6.023 \times 10^{23} \text{ Da/g}} \times M(\text{Da})$$

$$= 1.212 \times 10^{-3} (\text{nm}^3/\text{Da}) \times M(\text{Da}).$$

Assuming spherical shape $R_{\min} = (3V/4\pi)^{1/3}$

$$= 0.066 M^{1/3} (\text{for } M \text{ in Dalton, } R_{\min} \text{ in nanometer}).$$

Protein <i>M</i> (kDa)	5	10	20	50	100	200	500
<i>R</i> _{min} (nm)	1.1	1.42	1.78	2.4	3.05	3.84	5.21

Find the relation between average distance (nm) between molecules and its concentration (M)

In a 1-M solution, there are 6×10^{23} molecules/l, = 0.6 molecules/nm³, or inverting, the volume per molecule is $V = 1.66 \text{ nm}^3/\text{molecule}$ at 1 M. For a concentration C , the volume per molecule is $V = 1.66/C$.

We will take the cube root of the volume per molecule as an indication of the average separation.

$$d = V^{1/3} = 1.18/C^{1/3}, \quad [3.1]$$

where C is in molar and d is in nanometer. Table 2 gives some typical values.

Concentration	1 M	1 mM	1 μ M	1 nM
Distance between molecules (nm)	1.18	11.8	118	1,180

Controlled release: Basic kinetics

Rate of drug release from a polymeric device (matrix, particles etc.) can be approximated by:

a. Zero order release: Rate of drug release is independent of the mass of drug remaining in the device

E.g.: Osmotic and mechanical pumps, Rate-controlling membranes reservoir controlled release, surface-eroding polymers

b. First order release: Rate of drug release is proportional to the amount of drug remaining within the implant

E.g.: Bulk eroding polymeric devices

Controlled release: Basic kinetics

Zero order kinetics:

Rate of release is constant

$$\frac{dM_t}{dt} = k_o$$
$$M_t = k_o t$$

First order kinetics:

Rate of release is proportional to the remaining drug

$$\frac{dM_t}{dt} = k(M_\infty - M_t)$$
$$M_t = M_\infty [1 - \exp^{-kt}]$$
$$\frac{dM_t}{dt} = kM_\infty \exp^{-kt}$$